UNIVERSOPEDIA

Formal Observational Language Standard and Cross-Scale Physical Ledger

Complete Standalone Edition · 2026

CONTROLLING PRINCIPLE

A physical description is not a measurement until a mechanism identifies what quantity exists, how it couples to a channel, what observable follows, and what record is preserved. Universopedia therefore begins with records, preserves established inference separately, and adds UM reconstruction only as a labelled layer.

Prepared for Joseph Shields and for integration into the Unified Mechanics master document

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Authority, purpose, and status

THE ONE-SENTENCE DEFINITION

Universopedia is an open-world, source-traceable ledger of how physical systems become describable through mechanisms, channels, records, established inference, bounded reconstruction, and conservative familiarity classes.

This document is the dedicated formal statement of the Universopedia layer of Unified Mechanics. It consolidates the class dictionary, observational mathematics, periodic language family, cross-scale object ledgers, complexity profiles, cosmology evidence matrix, TNG test programme, TNG literature phase, SPARC pilot, mechanism-before-measurement rule, and governance standards into one self-contained reference.

It does not replace the structural UM monographs, nor does it convert the Universopedia into a new force law. Its role is to state exactly what a record is, what an instrument or natural channel preserved, what established science infers from the record, what UM may reconstruct, what was physically lost, and what remains recorded but unfamiliar.

R	Calibrated physical record.
E	Established inference under a declared model, channel, calibration, and uncertainty.
U	UM reconstruction or interpretive class layered on top of the established record and inference.
T	Transmission limit: distinctions the selected channel did not preserve.
F	Familiarity gap: recorded structure not yet assigned to a validated language class.

Document status

Component	Status	Meaning
Formal definitions and schemas	Closed for this edition	Definitions, maps, kernels, clocks, profiles, validation rules.
Periodic language family	Complete inventory	All 118 elements retained as entries and categorized.
Cross-scale atlas	Complete representative release	8 planets, 6 stellar classes, 5 galaxy classes, 4 black-hole cases.
Observable mathematics	Complete release	22 record-to-inference equations and validity boundaries.
Complexity profiles	Complete release	28 explicit scale-relative profiles.

Component	Status	Meaning
Cosmology comparison	Coverage complete; distinctiveness open	29 evidence rows, tension register, six comparison gates.
TNG object-level test	Deferred, not cancelled	Frozen protocol exists; exact adapter source and authenticated catalogue run remain outstanding.
TNG literature phase	Complete retrospective comparison	26 statements from 9 primary papers; language survives, superiority untested.
SPARC pilot	Executed real-data null/equivalence result	UM quadratic coordinates did not beat RAR and matched a generic quadratic basis.

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PART I — SCIENTIFIC CONTRACT

1. Mechanism must precede measurement

NON-NEGOTIABLE RULE

No mechanism → no derived observable → no physical prediction. A measurement can exist only after the framework specifies what is being measured and how it becomes a record.

physical world → mechanism → state variable → coupling
→ observable → instrument interaction → record → calibration → inference

The physical world is not presented as a complete table. A telescope records photons; an interferometer records phase or strain; a collider detector records tracks, timing, and deposited energy; a simulation catalogue stores selected derived fields. Each record exists because a specific mechanism and channel selected it. Universopedia is therefore a discipline for stating the selection chain rather than pretending that an image, spectrum, parameter, or catalogue row is the whole object.

1.1 Measurement contract

Required element	Question
Mechanism	What physical process occurs?
State variable	What quantity is produced, changed, transported, or retained?
Coupling	How does the quantity enter a measurable channel?
Observable	What numerical or categorical quantity follows?
Instrument interaction	What apparatus or natural process responds?
Record	What is actually stored: count, spectrum, image, strain, track, time series, field, or catalogue value?
Calibration	How are dimensions, units, selection effects, and uncertainties assigned?
Decoder/inference	What model maps the record into a familiar physical quantity or class?
Comparator	What established description predicts the same record?
Failure condition	What outcome opposes the mechanism rather than merely failing to favour it?

1.2 The whole physical world is not one observable

A theory may be universal in scope while every actual observation remains local, channel-limited, and scale-dependent. No instrument records every microscopic degree of freedom, every causal history, every interior state, and every relation at once. Scientific description succeeds by preserving sufficient invariants, stable statistics, and reproducible classes—not by making the entire universe into one raw dataset.

The correct demand is therefore not “measure everything.” It is: specify the smallest sufficient mechanism and record that can distinguish the proposed explanation from generic description.

1.3 Claim hierarchy

Level	Name	Establishes	Does not yet establish
1	Structural identity	An equality, closure, normal form, quotient, or invariant.	Physical realization.
2	Representation theorem	A canonical representation under declared hypotheses.	That Nature uses the representation.
3	Physical dictionary	Formal components receive physical names or roles.	Correct mechanism or dynamics.
4	Physical mechanism	A process law specifies evolution, coupling, exchange, or registration.	A measurable result without an observable and calibration.
5	Derived observable	A definite quantity follows from the mechanism.	That an instrument records it as assumed.
6	Measurement model	The observable is linked to channel, detector, record, and inference.	Agreement with data.
7	Prediction	A value, relation, distribution, or scaling is frozen before target inspection.	Empirical support.
8	Empirical test	Independent records and fair baselines are used.	Universal validation beyond tested scope.

1.4 Claim grades used throughout

Grade	Meaning	Editorial rule
D	Definition or notation	Meaning fixed by declaration.
T	Theorem under stated assumptions	Deductive consequence within a formal system.
E	Established external science	Accepted physical result or standard model relation.
R	Calibrated record	Observed or simulated data product.
U	UM interpretation or reconstruction	A proposed language mapping; not automatically new physics.
H	Optional new-dynamics hypothesis	A separately falsifiable physical candidate.
Q	Quarantined legacy claim	Historical material not governing until rederived and tested.

1.5 The three recurrent traps

- Calling an existing observation a UM prediction merely because the observation can be rewritten in UM notation.
- Demanding that an interpretive compiler beat a specialist empirical relation before UM has derived the same physical observable from its own mechanism.

- Treating lack of access to a huge archive as a conceptual failure when a reduced catalogue, hosted query, summary statistic, or smaller open dataset can execute the actual test.

PART II — FORMAL UNIVERSOPEDIA SYSTEM

2. Primitive typed spaces

Symbol	Type	Meaning
X_S	World-state space at declared scale S	Physical configurations relevant to the selected identity class.
C_S	Channel family	Available causal routes: electromagnetic, dynamical, gravitational-wave, particle, in-situ, timing, etc.
D_{S,c}	Record space	Calibrated data objects produced by channel c.
L_S	Validated language space	Stable classes and coordinates accepted for the declared scope.
Θ_S	Inference parameter space	Model-dependent physical quantities with units and uncertainty.
E_S	Environment space	Density, temperature, composition, symmetry state, background field, and boundary conditions.
T_S	Clock space	Signal, formation, material, and processing ages.

2.1 Mechanism, realization, record, and decoder

$$
\begin{aligned}
&P_{S,c}: X_S \times E_S \times T_S \to X_S && \text{[physical mechanism]}\\
&M_{S,c}: X_S \to D_{S,c} && \text{[measurement / realization map]}\\
&C_{S,c}: D_{S,c} \to L_S \cup \{\text{unfamiliar}\} && \text{[decoder / familiarization map]}\\
&I_{S,c,m}: D_{S,c} \to \Theta_S \times \text{Uncertainty} && \text{[model-dependent inference]}
\end{aligned}
$$

P determines what physical process is claimed. M includes the coupling, propagation, instrument response, selection effects, and noise that generate a record. C assigns preserved records to validated classes. I reconstructs physical parameters under an explicit model m. The same data object may support multiple inferences, and different channels may support different language classes for the same target.

2.2 Record–inference–reconstruction separation

$$
\begin{aligned}
R &= d \in D_{S,c}\\
E &= I_{S,c,m}(d)\\
U &= U_S(R, E, \text{Boundary, clocks, kernels, ensemble context})
\end{aligned}
$$

Universopedia requires separate columns or paragraphs for R, E, and U. A mass inferred from orbital dynamics is not a direct image. A temperature fitted from a spectrum is not a raw pixel value. A UM Boundary or retention interpretation is not established merely because it can be attached to the same record.

2.3 Actual Boundary

$$B = B(S, c, \Delta t, \text{identity criterion}, \text{environment})$$

The actual Boundary is the scale-indexed causal cut separating retained identity from exchanged information for a selected channel and timescale. It is not one universal shell. The same object may possess optical, chemical, nuclear, thermodynamic, gravitational, orbital, group, and causal-horizon Boundaries. A nominal geometric surface is only one possible proxy.

- A planet has atmosphere, surface, interior, magnetic, orbital, and gravitational Boundaries.
- A galaxy has disk, gas, halo, satellite, circumgalactic, group, and cluster Boundaries.
- A black-hole system has emitting-plasma, dynamical compact-object, and relativistic causal Boundaries.
- An instrument adds aperture, sensitivity, cadence, spectral-band, and reconstruction Boundaries.

2.4 Transmission Kernel

$$K^T_{S,c}(X) = \{\Delta X : M_{S,c}(X + \Delta X) = M_{S,c}(X)\}$$

K^T contains world-state distinctions erased by propagation, coupling, resolution, projection, opacity, selection, or instrument response. In the linear case it is the algebraic kernel of M. No more intelligent decoder can recover a distinction that never altered the record.

2.5 Familiarity Kernel

$$K^F_{S,c} = \{d \in D_{S,c} : C_{S,c}(d) = \text{unfamiliar}\}$$

K^F contains structure that is present in the record but not yet assigned to a validated language class. It is not generally algebraic. It can shrink through conservative language growth, improved models, new reference libraries, or verified pattern classes. Confusing K^F with K^T turns an unknown signal into a falsely inaccessible one.

2.6 Joint-channel refinement

$$K^T_{\text{joint}} = \bigcap_i K^T_i$$

Exact as a linear-kernel statement; shared systematics and nonlinear selection effects must still be audited.

Independent channels shrink the common blind region. Spectroscopy, timing, gravity, lensing, microwaves, neutrinos, gravitational waves, in-situ particles, and seismology reveal different aspects of one target. Adding a synonym does not shrink K^T; adding a genuinely informative physical channel can.

2.7 Four age clocks

$$\tau = (\tau\_signal, \tau\_formation, \tau\_material, \tau\_processing)$$

Clock	Definition	Example
Signal age	Propagation delay from event or emission to record.	A Jupiter image is tens of light-minutes old.
Formation age	Time since assembly of the selected effective object.	Jupiter formed about 4.5–4.6 Gyr ago.
Material age	Age distribution of constituent matter or nuclei.	Many nuclei predate the Solar System.
Processing age	Time since the last major local reconfiguration.	Atmospheric circulation and chemistry are active now.

Age statements must name both the identity class and the clock. Collapsing these four quantities into one number destroys information.

2.8 Effective objects and identity

$$UM\_\ell = (X\_\ell, T\_\ell, \pi\_\ell, G\_\ell, A\_\ell, C\_\ell, \tau\_\ell, E\_\ell)$$

An effective object is a metastable equivalence class at scale ℓ. Its state space, dynamics, coarse-graining, gradients, actions, channels, clocks, and environment are declared. Identity persists when an explicit criterion remains stable over a declared retention timescale; it does not require absolute isolation.

2.9 From local records to ensemble fields

$$F\_\ell(x) = [\Sigma\_i\ W\_\ell(x - x\_i)\ f(z\_i)]\ /\ [\Sigma\_i\ W\_\ell(x - x\_i)]$$
$$N\_eff = (\Sigma\_i\ w\_i)\hat{}2\ /\ \Sigma\_i\ w\_i\hat{}2$$

At N_eff near one, F_ℓ is effectively a local-object value. At large N_eff, and when the result is stable under justified resolution changes, field language becomes operationally legitimate. Density, temperature, pressure, velocity, composition, radiation intensity, gradients, and correlations are retained ensemble descriptions—not extra substances added to the microscopic contributors.

2.10 Complexity profile

$$\kappa = (\text{Diversity, Nesting, Retention, Channels, ModelDepth, Uncertainty})$$

Universopedia rejects a single universal scalar essence of complexity. A system may be more complex on one component and less on another. The Navigation Index averages the first five ordinal scores for sorting only; uncertainty remains separate and the index is not a law of nature.

2.11 Structural channel grammar

$$
\begin{gathered}
a\_L + a\_M = 1 \\
W\_L = a\_L^2 \\
W\_B = 2\, a\_L\, a\_M \\
W\_M = a\_M^2 \\
W\_L + W\_B + W\_M = 1
\end{gathered}
$$

The quadratic triad is an exact algebraic decomposition after the local amplitudes are declared. It can encode availability/update, relation/boundary, and retention/matter-like roles. It is not automatically a cosmological density partition, force law, or measured coupling. The SPARC pilot showed that a linear model in W_B and W_M is algebraically equivalent to a generic quadratic model in a_L when an intercept is present.

2.12 Golden structural coordinates

$$
\begin{gathered}
r = (\sqrt{5} - 1)/4 \approx 0.3090169944 \\
u = 1 - r \approx 0.6909830056 \\
4r^2 + 2r - 1 = 0
\end{gathered}
$$

These are structural coordinates selected by the declared recursion and normalization. Universopedia may use them as internal coordinates, but it must not promote them to densities, ages, couplings, or observational values by numerical resemblance alone.

2.13 Channel-weighted inference module

U	Proposed observational module; physical response weights must be derived or calibrated for each probe.

$$
\begin{gathered}
w\_p = (\ell\_p, b\_p, m\_p), \qquad \ell\_p + b\_p + m\_p = 1 \\
O\_p = \ell\_p\, O\_L + b\_p\, O\_B + m\_p\, O\_M \\
\hat{\theta}\_p = w\_p \cdot \theta \\
\Sigma\_\{pq\} = (w\_p - w\_q) \cdot \theta
\end{gathered}
$$

Probe disagreement can be represented as a difference between channel-response vectors rather than as a claim that one probe sees the whole object. A proposed response-Gram construction is:

$$
\begin{gathered}
L\_p = \langle j\_U, j\_U \rangle \\
B\_p = 2\langle j\_U, j\_C \rangle \\
M\_p = \langle j\_C, j\_C \rangle \\
(\ell\_p,b\_p,m\_p) = (L\_p,B\_p,M\_p)/(L\_p+B\_p+M\_p)
\end{gathered}
$$

This module becomes physical only after j_U and j_C are linked to an actual mechanism and detector response.

PART III — LEDGER GRAMMAR AND VALIDATION

3. Core class dictionary

ID	Term	Operational definition	Form	Layer	Guardrail
LC-001	Physical channel	A causal route by which a system leaves a record.	Photons, neutrinos, gravitational waves, particle samples, fields, timing.	R	Specify instrument, band and calibration.
LC-002	Record	The calibrated data object produced by a channel.	Image pixels, spectrum, light curve, interferometric visibilities.	R	Do not call an inferred mass a direct observation.
LC-003	Measurement map M_S	The physical and instrument map from world-state X to record D for scale S.	X → D	R/E	Include selection effects and noise.
LC-004	Decoder C_S	A rule assigning a record to a validated language class or an unresolved state.	D → L union {unfamiliar}	E	A decoder is model-dependent.
LC-005	Transmission Kernel K^T	World-state differences erased or never recorded by the chosen channel.	K^T=ker(M) in the linear case.	T	Cannot be recovered from the identical record by a cleverer decoder.
LC-006	Familiarity Kernel K^F	Records containing structure the current validated language cannot classify.	K^F={d:C(d)=unfamiliar}	F	Not generally algebraic; can shrink through conservative language growth.
LC-007	Joint channel	Independent measurement maps applied to the same target.	ker(M_joint)=intersection of channel kernels in the linear case.	E/U	Shared systematics must be checked.
LC-008	Actual Boundary	A scale-indexed causal cut separating retained identity from exchanged information.	Depends on scale, channel and timescale.	U	It is not one universal physical surface.
LC-009	Identity criterion	The observable condition under which an object is treated as the same effective object.	Metastability over a declared timescale.	E/U	State the retention timescale.
LC-010	Signal age	Travel time between emission and reception.	Distance / propagation speed when geometry permits.	E	Not the object's formation age.
LC-011	Formation age	Elapsed time since an effective object formed.	Model-dependent chronometry.	E	Often a range.
LC-012	Material age	Age distribution of constituent matter or nuclei.	Nucleosynthetic and geochemical clocks.	E	Often older than the assembled object.
LC-013	Processing age	Time since matter was last substantially reconfigured.	Thermal, chemical, impact, accretion or merger clock.	E	Regions can have different clocks.
LC-014	Familiar language class	A validated class mapping records to a stable observational meaning.	Hydrogen line, gas giant, main-sequence star, lensing halo.	E/U	Class expansion must preserve prior successes.
LC-015	Environment gate	A local condition activating or suppressing an effective response.	Density, temperature, composition, symmetry state.	E/U	A UM scalar gate remains hypothetical until independently tested.
LC-016	Complexity profile	A vector: diversity, nesting, retention, channels, model depth and uncertainty.	kappa=(D,N,R,C,M,U)	U	Descriptive and scale-relative.
LC-017	Navigation index	Average of declared complexity-profile scores.	mean(D,N,R,C,M)	U	For sorting only; not an ontological law.

ID	Term	Operational definition	Form	Layer	Guardrail
LC-018	Claim grade	Label stating the epistemic layer of a cell or row.	R, E, U, T, F.	R/E/U	Mixed rows must keep layers in separate columns.

3.1 Standard entry schema

Field	Required content
Entry_ID	Stable versioned identifier.
Entity/Object	Human-readable target name.
Class	Validated or provisional language class.
Scale	Spatial, temporal, energy, or resolution scale.
Identity criterion	Condition under which the effective object is treated as the same object.
Mechanism	Physical process responsible for the observable when a physical prediction is claimed.
Direct records (R)	Calibrated channel outputs only.
Established inference (E)	Model, units, uncertainty, and validity boundary.
Actual Boundary	Scale/channel/timescale-dependent causal cut.
Dominant channels	Instrument, band, messenger, cadence, geometry, and calibration.
Transmission limit (T)	Distinctions absent from the record.
Familiarity class/gap (F)	Validated class or unresolved recorded structure.
Four clocks	Signal, formation, material, processing.
Identity retention	Why the object/class persists over the declared timescale.
Complexity profile	κ vector and evidence note.
UM reconstruction (U)	Interpretive map, explicitly separated from R and E.
Claim grade	D/T/E/R/U/H/Q.
Source and access date	Primary or official reference and version.
Validation status	Completeness, layer separation, source traceability, test status.

3.2 Physical channel taxonomy

Channel family	Typical records	Typical Transmission limit
Imaging / photometry	Pixels, fluxes, colours, morphology, time series	Depth, interior state, composition degeneracy, projection.
Spectroscopy	Line positions, widths, ratios, continua, polarization	Opacity, spectral coverage, blending, non-unique plasma/abundance models.
Timing / astrometry	Periods, delays, positions, proper motions, parallax	Composition and most microscopic histories.

Channel family	Typical records	Typical Transmission limit
Dynamics / gravity	Orbits, accelerations, velocity dispersions, tides	Detailed composition and non-gravitating state variables.
Gravitational lensing	Image distortions, shear, convergence, time delays	Mass-sheet and line-of-sight degeneracies; composition.
Gravitational waves	Strain time series and phase evolution	Electromagnetic composition and many environmental details.
Particle / neutrino	Counts, energies, tracks, flavours, arrival directions	Uncoupled sectors, source degeneracy, incomplete interaction history.
In-situ sampling	Local particles, chemistry, fields, temperature, pressure	Global structure beyond the sampled path or region.
Seismic / acoustic	Travel times, mode frequencies, attenuation	Fine structure below resolution and model-dependent interior degeneracy.
Magnetic / electric field	Local or remote field components, Faraday rotation	Current topology and source distribution can remain non-unique.
Simulation / numerical catalogue	State snapshots and derived fields under a declared solver	Nature beyond model assumptions; unresolved scales; calibration to observation.

3.3 Formal consequences

Consequence	Statement
T1 — Record/inference separation	If two inference models map one record to different parameters, the record remains the same; inferred parameters are not direct observations.
T2 — Transmission impossibility	If ΔX lies in K^T, no decoder acting on the identical record can recover that distinction.
T3 — Familiarity reducibility	A record in K^F may become familiar without new physical data if a conservative validated class is learned.
T4 — Joint-channel refinement	Independent informative channels can reduce the common Transmission Kernel.
T5 — Boundary relativity	Changing scale, channel, timescale, or identity criterion can change the actual Boundary without changing the underlying target.
T6 — Age non-collapse	Signal, formation, material, and processing ages are generally non-identical and cannot be replaced by one age without loss.
T7 — Compiler non-creation	A deterministic compiler cannot create target information absent from its inputs; distinctiveness requires transfer or withheld prediction against information-matched baselines.
T8 — Open-world conservativity	A new familiarity class is admissible only if it preserves previously validated distinctions and improves traceable classification.

3.4 Validation gates

Gate	Question	Pass rule	Current status
G1 - Coverage	Can each probe keep R, E and U separate and name its Boundary?	All audited rows complete with no silent layer crossing.	PASS
G2 - Resolution stability	Do important labels persist when justified resolution changes?	Same frozen rule within declared tolerance.	PARTIAL
G3 - Transfer	Does one fixed compiler work across instruments/systems?	No retuning between declared training and transfer sets.	PARTIAL
G4 - Withheld prediction	Does the compiler predict records not used to build it?	Prospective held-out improvement with uncertainty and covariance.	NOT EXECUTED
G5 - Generic baseline advantage	Does UM beat simpler density/mixing/clustering descriptions?	Pre-registered improvement after complexity penalty.	NOT SHOWN
G6 - Optional scalar likelihood	Does the fixed response ratio survive all existing constraints?	Stable solution, joint viable region, then held-out fit beats free-ratio penalty.	NOT EXECUTED

Coverage is the first gate, not the last. A ledger can be complete, internally consistent, and scientifically useful while still providing no independent prediction or baseline advantage.

3.5 Failure modes prohibited by the standard

- Replacing a record with its interpretation in prose or tables.
- Treating an unknown recorded pattern as physically absent.
- Treating an absent distinction as recoverable through a better synonym or decoder.
- Using one universal Boundary, one universal observer, one universal clock, or one scalar complexity score.
- Promoting numerical resemblance to derivation or measurement.
- Using target variables to construct the adapter intended to predict them.
- Retuning a compiler after inspecting transfer targets.
- Counting a successful reinterpretation of an accepted simulation as evidence for altered physical dynamics.
- Hiding null results, source-model tensions, or baseline equivalence.
- Calling a structural coordinate a measured density or coupling without a physical mechanism and calibration.

PART IV — THE PERIODIC TABLE AS THE FIRST COMPLETE UNIVERSOPEDIA

4. Why the periodic table is the first complete language family

Atomic identities recur across laboratories, planets, stars, interstellar media, and engineered systems. Laboratory line identities, wavelength ratios, isotope shifts, selection rules, and mass records make remote matter familiar through repeatable classes. The periodic table therefore provides the first complete physical language family: every named element has a stable proton-number identity, environment-dependent electronic subclasses, measurable transitions, and explicit limits on what one channel preserves.

$$\Gamma\_(Z,A,e) = (E\_levels, \sigma\_abs, \sigma\_emit, \tau\_levels, B\_nuc, \alpha\_pol, phase)$$

Z specifies nuclear charge, A isotope, and e environment/ionization/bonding context. The tuple is not a claim that one spectrum reveals every component. It is a schema for the family of records and inferences associated with an atomic identity class.

4.1 Atomic identity hierarchy

Layer	Identity refinement
Element	Fixed proton number Z.
Isotope	Fixed Z and mass number A.
Ionization state	Electron number and charge class.
Electronic state	Energy-level and transition class.
Molecular / bonding environment	Local chemistry and symmetry.
Phase / material environment	Gas, plasma, liquid, solid, lattice, pressure-temperature state.
Nucleosynthetic provenance	Primordial, stellar, explosive, spallation, or synthetic origin history.

4.2 Categorization summary

Category	Class	Entries
Family	Transition metal	38
Family	Actinide	15
Family	Lanthanide	15
Family	Post-transition metal	8
Family	Noble gas	7

Category	Class	Entries
Family	Reactive nonmetal	7
Family	Alkali metal	6
Family	Alkaline-earth metal	6
Family	Halogen	6
Family	Metalloid (conventional)	6
Family	Superheavy; chemistry partly predicted	4
Block	d	38
Block	f	30
Block	p	36
Block	s	14
Period	1	2
Period	2	8
Period	3	8
Period	4	18
Period	5	18
Period	6	32
Period	7	32

4.3 Guardrails

- A line identification does not alone determine abundance, isotope history, chemical environment, or interior distribution.
- Element classes do not license element-specific gravity or arbitrary new forces.
- Opacity, resolution, blending, selection rules, and missing spectral bands belong to the Transmission Kernel.
- A recorded unidentified line belongs to the Familiarity Kernel until a conservative classification is established.
- Remote abundance and temperature are ensemble inferences from line families, continua, transfer models, and calibration—not properties extracted from one isolated atom.

4.4 Full 118-element categorized catalogue

Z	Sym	Element	Period	Group/series	Block	Family	Atomic weight / mass	Isotope/provenance	Origin clock	UM use
1	H	Hydrogen	1	Group 1	s	Reactive nonmetal	1.0080	At least one stable isotope	Big-Bang nucleosynthesis, then stellar processing	Atomic boundary class; relates nuclear identity, electron-shell response and environment-dependent chemistry.
2	He	Helium	1	Group 18	s	Noble gas	4.0026	At least one stable isotope	Big-Bang nucleosynthesis, then stellar processing	Closed-shell familiarity class; tests low-reactivity identity retention.
3	Li	Lithium	2	Group 1	s	Alkali metal	6.94	At least one stable isotope	Primordial trace, cosmic-ray spallation and stellar channels	Valence-boundary class; tests high exchange propensity.
4	Be	Beryllium	2	Group 2	s	Alkaline-earth metal	9.0122	At least one stable isotope	Primordial trace, cosmic-ray spallation and stellar channels	Valence-boundary class; tests high exchange propensity.
5	B	Boron	2	Group 13	p	Metalloid (conventional)	10.81	At least one stable isotope	Primordial trace, cosmic-ray spallation and stellar channels	Atomic boundary class; relates nuclear identity, electron-shell response and environment-dependent chemistry.
6	C	Carbon	2	Group 14	p	Reactive nonmetal	12.011	At least one stable isotope	Stellar fusion and explosive nucleosynthesis	Atomic boundary class; relates nuclear identity, electron-shell response and environment-dependent chemistry.
7	N	Nitrogen	2	Group 15	p	Reactive nonmetal	14.007	At least one stable isotope	Stellar fusion and explosive nucleosynthesis	Atomic boundary class; relates nuclear identity, electron-shell response and environment-dependent chemistry.
8	O	Oxygen	2	Group 16	p	Reactive nonmetal	15.999	At least one stable isotope	Stellar fusion and explosive nucleosynthesis	Atomic boundary class; relates nuclear identity, electron-shell response and environment-dependent chemistry.
9	F	Fluorine	2	Group 17	p	Halogen	18.998	At least one stable isotope	Stellar fusion and explosive nucleosynthesis	Valence-boundary class; tests high exchange propensity.
10	Ne	Neon	2	Group 18	p	Noble gas	20.180	At least one stable isotope	Stellar fusion and explosive nucleosynthesis	Closed-shell familiarity class; tests low-reactivity identity retention.
11	Na	Sodium	3	Group 1	s	Alkali metal	22.990	At least one stable isotope	Stellar fusion and explosive nucleosynthesis	Valence-boundary class; tests high exchange propensity.
12	Mg	Magnesium	3	Group 2	s	Alkaline-earth metal	24.305	At least one stable isotope	Stellar fusion and explosive nucleosynthesis	Valence-boundary class; tests high exchange propensity.
13	Al	Aluminium	3	Group 13	p	Post-transition metal	26.982	At least one stable isotope	Stellar fusion and explosive nucleosynthesis	Atomic boundary class; relates nuclear identity, electron-shell response and environment-dependent chemistry.
14	Si	Silicon	3	Group 14	p	Metalloid (conventional)	28.085	At least one stable isotope	Stellar fusion and explosive nucleosynthesis	Atomic boundary class; relates nuclear identity, electron-shell response and environment-dependent chemistry.

Z	Sym	Element	Period	Group/series	Block	Family	Atomic weight / mass	Isotope/provenance	Origin clock	UM use
15	P	Phosphorus	3	Group 15	p	Reactive nonmetal	30.974	At least one stable isotope	Stellar fusion and explosive nucleosynthesis	Atomic boundary class; relates nuclear identity, electron-shell response and environment-dependent chemistry.
16	S	Sulfur	3	Group 16	p	Reactive nonmetal	32.06	At least one stable isotope	Stellar fusion and explosive nucleosynthesis	Atomic boundary class; relates nuclear identity, electron-shell response and environment-dependent chemistry.
17	Cl	Chlorine	3	Group 17	p	Halogen	35.45	At least one stable isotope	Stellar fusion and explosive nucleosynthesis	Valence-boundary class; tests high exchange propensity.
18	Ar	Argon	3	Group 18	p	Noble gas	39.95	At least one stable isotope	Stellar fusion and explosive nucleosynthesis	Closed-shell familiarity class; tests low-reactivity identity retention.
19	K	Potassium	4	Group 1	s	Alkali metal	39.098	At least one stable isotope	Stellar fusion and explosive nucleosynthesis	Valence-boundary class; tests high exchange propensity.
20	Ca	Calcium	4	Group 2	s	Alkaline-earth metal	40.078	At least one stable isotope	Stellar fusion and explosive nucleosynthesis	Valence-boundary class; tests high exchange propensity.
21	Sc	Scandium	4	Group 3	d	Transition metal	44.956	At least one stable isotope	Stellar fusion and explosive nucleosynthesis	Atomic boundary class; relates nuclear identity, electron-shell response and environment-dependent chemistry.
22	Ti	Titanium	4	Group 4	d	Transition metal	47.867	At least one stable isotope	Stellar fusion and explosive nucleosynthesis	Atomic boundary class; relates nuclear identity, electron-shell response and environment-dependent chemistry.
23	V	Vanadium	4	Group 5	d	Transition metal	50.942	At least one stable isotope	Stellar fusion and explosive nucleosynthesis	Atomic boundary class; relates nuclear identity, electron-shell response and environment-dependent chemistry.
24	Cr	Chromium	4	Group 6	d	Transition metal	51.996	At least one stable isotope	Stellar fusion and explosive nucleosynthesis	Atomic boundary class; relates nuclear identity, electron-shell response and environment-dependent chemistry.
25	Mn	Manganese	4	Group 7	d	Transition metal	54.938	At least one stable isotope	Stellar fusion and explosive nucleosynthesis	Atomic boundary class; relates nuclear identity, electron-shell response and environment-dependent chemistry.
26	Fe	Iron	4	Group 8	d	Transition metal	55.845	At least one stable isotope	Stellar fusion and explosive nucleosynthesis	Atomic boundary class; relates nuclear identity, electron-shell response and environment-dependent chemistry.
27	Co	Cobalt	4	Group 9	d	Transition metal	58.933	At least one stable isotope	Explosive nucleosynthesis and neutron-capture families	Atomic boundary class; relates nuclear identity, electron-shell response and environment-dependent chemistry.
28	Ni	Nickel	4	Group 10	d	Transition metal	58.693	At least one stable isotope	Explosive nucleosynthesis and neutron-capture families	Atomic boundary class; relates nuclear identity, electron-shell response and environment-dependent chemistry.

Z	Sym	Element	Period	Group/series	Block	Family	Atomic weight / mass	Isotope/provenance	Origin clock	UM use
29	Cu	Copper	4	Group 11	d	Transition metal	63.546	At least one stable isotope	Explosive nucleosynthesis and neutron-capture families	Atomic boundary class; relates nuclear identity, electron-shell response and environment-dependent chemistry.
30	Zn	Zinc	4	Group 12	d	Transition metal	65.38	At least one stable isotope	Explosive nucleosynthesis and neutron-capture families	Atomic boundary class; relates nuclear identity, electron-shell response and environment-dependent chemistry.
31	Ga	Gallium	4	Group 13	p	Post-transition metal	69.723	At least one stable isotope	Explosive nucleosynthesis and neutron-capture families	Atomic boundary class; relates nuclear identity, electron-shell response and environment-dependent chemistry.
32	Ge	Germanium	4	Group 14	p	Metalloid (conventional)	72.630	At least one stable isotope	Explosive nucleosynthesis and neutron-capture families	Atomic boundary class; relates nuclear identity, electron-shell response and environment-dependent chemistry.
33	As	Arsenic	4	Group 15	p	Metalloid (conventional)	74.922	At least one stable isotope	Explosive nucleosynthesis and neutron-capture families	Atomic boundary class; relates nuclear identity, electron-shell response and environment-dependent chemistry.
34	Se	Selenium	4	Group 16	p	Reactive nonmetal	78.971	At least one stable isotope	Explosive nucleosynthesis and neutron-capture families	Atomic boundary class; relates nuclear identity, electron-shell response and environment-dependent chemistry.
35	Br	Bromine	4	Group 17	p	Halogen	79.904	At least one stable isotope	Explosive nucleosynthesis and neutron-capture families	Valence-boundary class; tests high exchange propensity.
36	Kr	Krypton	4	Group 18	p	Noble gas	83.798	At least one stable isotope	Explosive nucleosynthesis and neutron-capture families	Closed-shell familiarity class; tests low-reactivity identity retention.
37	Rb	Rubidium	5	Group 1	s	Alkali metal	85.468	At least one stable isotope	Explosive nucleosynthesis and neutron-capture families	Valence-boundary class; tests high exchange propensity.
38	Sr	Strontium	5	Group 2	s	Alkaline-earth metal	87.62	At least one stable isotope	Explosive nucleosynthesis and neutron-capture families	Valence-boundary class; tests high exchange propensity.
39	Y	Yttrium	5	Group 3	d	Transition metal	88.906	At least one stable isotope	Explosive nucleosynthesis and neutron-capture families	Atomic boundary class; relates nuclear identity, electron-shell response and environment-dependent chemistry.
40	Zr	Zirconium	5	Group 4	d	Transition metal	91.224	At least one stable isotope	Explosive nucleosynthesis and neutron-capture families	Atomic boundary class; relates nuclear identity, electron-shell response and environment-dependent chemistry.
41	Nb	Niobium	5	Group 5	d	Transition metal	92.906	At least one stable isotope	Explosive nucleosynthesis and neutron-capture families	Atomic boundary class; relates nuclear identity, electron-shell response and environment-dependent chemistry.
42	Mo	Molybdenum	5	Group 6	d	Transition metal	95.95	At least one stable isotope	Explosive nucleosynthesis and neutron-capture families	Atomic boundary class; relates nuclear identity, electron-shell response and environment-dependent chemistry.

Z	Sym	Element	Period	Group/series	Block	Family	Atomic weight / mass	Isotope/provenance	Origin clock	UM use
43	Tc	Technetium	5	Group 7	d	Transition metal	[97]	No stable isotope; trace natural	Explosive nucleosynthesis and neutron-capture families	Atomic boundary class; relates nuclear identity, electron-shell response and environment-dependent chemistry.
44	Ru	Ruthenium	5	Group 8	d	Transition metal	101.07	At least one stable isotope	Explosive nucleosynthesis and neutron-capture families	Atomic boundary class; relates nuclear identity, electron-shell response and environment-dependent chemistry.
45	Rh	Rhodium	5	Group 9	d	Transition metal	102.91	At least one stable isotope	Explosive nucleosynthesis and neutron-capture families	Atomic boundary class; relates nuclear identity, electron-shell response and environment-dependent chemistry.
46	Pd	Palladium	5	Group 10	d	Transition metal	106.42	At least one stable isotope	Explosive nucleosynthesis and neutron-capture families	Atomic boundary class; relates nuclear identity, electron-shell response and environment-dependent chemistry.
47	Ag	Silver	5	Group 11	d	Transition metal	107.87	At least one stable isotope	Explosive nucleosynthesis and neutron-capture families	Atomic boundary class; relates nuclear identity, electron-shell response and environment-dependent chemistry.
48	Cd	Cadmium	5	Group 12	d	Transition metal	112.41	At least one stable isotope	Explosive nucleosynthesis and neutron-capture families	Atomic boundary class; relates nuclear identity, electron-shell response and environment-dependent chemistry.
49	In	Indium	5	Group 13	p	Post-transition metal	114.82	At least one stable isotope	Explosive nucleosynthesis and neutron-capture families	Atomic boundary class; relates nuclear identity, electron-shell response and environment-dependent chemistry.
50	Sn	Tin	5	Group 14	p	Post-transition metal	118.71	At least one stable isotope	Explosive nucleosynthesis and neutron-capture families	Atomic boundary class; relates nuclear identity, electron-shell response and environment-dependent chemistry.
51	Sb	Antimony	5	Group 15	p	Metalloid (conventional)	121.76	At least one stable isotope	Explosive nucleosynthesis and neutron-capture families	Atomic boundary class; relates nuclear identity, electron-shell response and environment-dependent chemistry.
52	Te	Tellurium	5	Group 16	p	Metalloid (conventional)	127.60	At least one stable isotope	Explosive nucleosynthesis and neutron-capture families	Atomic boundary class; relates nuclear identity, electron-shell response and environment-dependent chemistry.
53	I	Iodine	5	Group 17	p	Halogen	126.90	At least one stable isotope	Explosive nucleosynthesis and neutron-capture families	Valence-boundary class; tests high exchange propensity.
54	Xe	Xenon	5	Group 18	p	Noble gas	131.29	At least one stable isotope	Explosive nucleosynthesis and neutron-capture families	Closed-shell familiarity class; tests low-reactivity identity retention.
55	Cs	Caesium	6	Group 1	s	Alkali metal	132.91	At least one stable isotope	Explosive nucleosynthesis and neutron-capture families	Valence-boundary class; tests high exchange propensity.
56	Ba	Barium	6	Group 2	s	Alkaline-earth metal	137.33	At least one stable isotope	Explosive nucleosynthesis and neutron-capture families	Valence-boundary class; tests high exchange propensity.

Z	Sym	Element	Period	Group/series	Block	Family	Atomic weight / mass	Isotope/provenance	Origin clock	UM use
57	La	Lanthanum	6	Lanthanide series	f	Lanthanide	138.91	At least one stable isotope	Explosive nucleosynthesis and neutron-capture families	Nested electron/nuclear channels; tests partial familiarity and radioactive clocks.
58	Ce	Cerium	6	Lanthanide series	f	Lanthanide	140.12	At least one stable isotope	Explosive nucleosynthesis and neutron-capture families	Nested electron/nuclear channels; tests partial familiarity and radioactive clocks.
59	Pr	Praseodymium	6	Lanthanide series	f	Lanthanide	140.91	At least one stable isotope	Explosive nucleosynthesis and neutron-capture families	Nested electron/nuclear channels; tests partial familiarity and radioactive clocks.
60	Nd	Neodymium	6	Lanthanide series	f	Lanthanide	144.24	At least one stable isotope	Explosive nucleosynthesis and neutron-capture families	Nested electron/nuclear channels; tests partial familiarity and radioactive clocks.
61	Pm	Promethium	6	Lanthanide series	f	Lanthanide	[145]	No stable isotope; trace natural	Explosive nucleosynthesis and neutron-capture families	Nested electron/nuclear channels; tests partial familiarity and radioactive clocks.
62	Sm	Samarium	6	Lanthanide series	f	Lanthanide	150.36	At least one stable isotope	Explosive nucleosynthesis and neutron-capture families	Nested electron/nuclear channels; tests partial familiarity and radioactive clocks.
63	Eu	Europium	6	Lanthanide series	f	Lanthanide	151.96	At least one stable isotope	Explosive nucleosynthesis and neutron-capture families	Nested electron/nuclear channels; tests partial familiarity and radioactive clocks.
64	Gd	Gadolinium	6	Lanthanide series	f	Lanthanide	157.25	At least one stable isotope	Explosive nucleosynthesis and neutron-capture families	Nested electron/nuclear channels; tests partial familiarity and radioactive clocks.
65	Tb	Terbium	6	Lanthanide series	f	Lanthanide	158.93	At least one stable isotope	Explosive nucleosynthesis and neutron-capture families	Nested electron/nuclear channels; tests partial familiarity and radioactive clocks.
66	Dy	Dysprosium	6	Lanthanide series	f	Lanthanide	162.50	At least one stable isotope	Explosive nucleosynthesis and neutron-capture families	Nested electron/nuclear channels; tests partial familiarity and radioactive clocks.
67	Ho	Holmium	6	Lanthanide series	f	Lanthanide	164.93	At least one stable isotope	Explosive nucleosynthesis and neutron-capture families	Nested electron/nuclear channels; tests partial familiarity and radioactive clocks.
68	Er	Erbium	6	Lanthanide series	f	Lanthanide	167.26	At least one stable isotope	Explosive nucleosynthesis and neutron-capture families	Nested electron/nuclear channels; tests partial familiarity and radioactive clocks.
69	Tm	Thulium	6	Lanthanide series	f	Lanthanide	168.93	At least one stable isotope	Explosive nucleosynthesis and neutron-capture families	Nested electron/nuclear channels; tests partial familiarity and radioactive clocks.

Z	Sym	Element	Period	Group/series	Block	Family	Atomic weight / mass	Isotope/provenance	Origin clock	UM use
70	Yb	Ytterbium	6	Lanthanide series	f	Lanthanide	173.05	At least one stable isotope	Explosive nucleosynthesis and neutron-capture families	Nested electron/nuclear channels; tests partial familiarity and radioactive clocks.
71	Lu	Lutetium	6	Lanthanide series	f	Lanthanide	174.97	At least one stable isotope	Explosive nucleosynthesis and neutron-capture families	Nested electron/nuclear channels; tests partial familiarity and radioactive clocks.
72	Hf	Hafnium	6	Group 4	d	Transition metal	178.49	At least one stable isotope	Explosive nucleosynthesis and neutron-capture families	Atomic boundary class; relates nuclear identity, electron-shell response and environment-dependent chemistry.
73	Ta	Tantalum	6	Group 5	d	Transition metal	180.95	At least one stable isotope	Explosive nucleosynthesis and neutron-capture families	Atomic boundary class; relates nuclear identity, electron-shell response and environment-dependent chemistry.
74	W	Tungsten	6	Group 6	d	Transition metal	183.84	At least one stable isotope	Explosive nucleosynthesis and neutron-capture families	Atomic boundary class; relates nuclear identity, electron-shell response and environment-dependent chemistry.
75	Re	Rhenium	6	Group 7	d	Transition metal	186.21	At least one stable isotope	Explosive nucleosynthesis and neutron-capture families	Atomic boundary class; relates nuclear identity, electron-shell response and environment-dependent chemistry.
76	Os	Osmium	6	Group 8	d	Transition metal	190.23	At least one stable isotope	Explosive nucleosynthesis and neutron-capture families	Atomic boundary class; relates nuclear identity, electron-shell response and environment-dependent chemistry.
77	Ir	Iridium	6	Group 9	d	Transition metal	192.22	At least one stable isotope	Explosive nucleosynthesis and neutron-capture families	Atomic boundary class; relates nuclear identity, electron-shell response and environment-dependent chemistry.
78	Pt	Platinum	6	Group 10	d	Transition metal	195.08	At least one stable isotope	Explosive nucleosynthesis and neutron-capture families	Atomic boundary class; relates nuclear identity, electron-shell response and environment-dependent chemistry.
79	Au	Gold	6	Group 11	d	Transition metal	196.97	At least one stable isotope	Explosive nucleosynthesis and neutron-capture families	Atomic boundary class; relates nuclear identity, electron-shell response and environment-dependent chemistry.
80	Hg	Mercury	6	Group 12	d	Transition metal	200.59	At least one stable isotope	Explosive nucleosynthesis and neutron-capture families	Atomic boundary class; relates nuclear identity, electron-shell response and environment-dependent chemistry.
81	Tl	Thallium	6	Group 13	p	Post-transition metal	204.38	At least one stable isotope	Explosive nucleosynthesis and neutron-capture families	Atomic boundary class; relates nuclear identity, electron-shell response and environment-dependent chemistry.
82	Pb	Lead	6	Group 14	p	Post-transition metal	207.2	At least one stable isotope	Explosive nucleosynthesis and neutron-capture families	Atomic boundary class; relates nuclear identity, electron-shell response and environment-dependent chemistry.

Z	Sym	Element	Period	Group/series	Block	Family	Atomic weight / mass	Isotope/provenance	Origin clock	UM use
83	Bi	Bismuth	6	Group 15	p	Post-transition metal	208.98	Primordial or trace radioactive	Explosive nucleosynthesis and neutron-capture families	Atomic boundary class; relates nuclear identity, electron-shell response and environment-dependent chemistry.
84	Po	Polonium	6	Group 16	p	Post-transition metal	[209]	Primordial or trace radioactive	Radioactive r-process or related heavy-element channels	Atomic boundary class; relates nuclear identity, electron-shell response and environment-dependent chemistry.
85	At	Astatine	6	Group 17	p	Halogen	[210]	Primordial or trace radioactive	Radioactive r-process or related heavy-element channels	Valence-boundary class; tests high exchange propensity.
86	Rn	Radon	6	Group 18	p	Noble gas	[222]	Primordial or trace radioactive	Radioactive r-process or related heavy-element channels	Closed-shell familiarity class; tests low-reactivity identity retention.
87	Fr	Francium	7	Group 1	s	Alkali metal	[223]	Primordial or trace radioactive	Radioactive r-process or related heavy-element channels	Valence-boundary class; tests high exchange propensity.
88	Ra	Radium	7	Group 2	s	Alkaline-earth metal	[226]	Primordial or trace radioactive	Radioactive r-process or related heavy-element channels	Valence-boundary class; tests high exchange propensity.
89	Ac	Actinium	7	Actinide series	f	Actinide	[227]	Primordial or trace radioactive	Radioactive r-process or related heavy-element channels	Nested electron/nuclear channels; tests partial familiarity and radioactive clocks.
90	Th	Thorium	7	Actinide series	f	Actinide	232.04	Primordial or trace radioactive	Radioactive r-process or related heavy-element channels	Nested electron/nuclear channels; tests partial familiarity and radioactive clocks.
91	Pa	Protactinium	7	Actinide series	f	Actinide	231.04	Primordial or trace radioactive	Radioactive r-process or related heavy-element channels	Nested electron/nuclear channels; tests partial familiarity and radioactive clocks.
92	U	Uranium	7	Actinide series	f	Actinide	238.03	Primordial or trace radioactive	Radioactive r-process or related heavy-element channels	Nested electron/nuclear channels; tests partial familiarity and radioactive clocks.
93	Np	Neptunium	7	Actinide series	f	Actinide	[237]	Trace natural; mostly synthetic	Laboratory synthesis; astrophysical occurrence not a normal abundance class	Nested electron/nuclear channels; tests partial familiarity and radioactive clocks.
94	Pu	Plutonium	7	Actinide series	f	Actinide	[244]	Trace natural; mostly synthetic	Laboratory synthesis; astrophysical occurrence not a normal abundance class	Nested electron/nuclear channels; tests partial familiarity and radioactive clocks.
95	Am	Americium	7	Actinide series	f	Actinide	[243]	Synthetic	Laboratory synthesis; astrophysical occurrence not a normal abundance class	Nested electron/nuclear channels; tests partial familiarity and radioactive clocks.
96	Cm	Curium	7	Actinide series	f	Actinide	[247]	Synthetic	Laboratory synthesis; astrophysical occurrence not a normal abundance class	Nested electron/nuclear channels; tests partial familiarity and radioactive clocks.

Z	Sym	Element	Period	Group/series	Block	Family	Atomic weight / mass	Isotope/provenance	Origin clock	UM use
97	Bk	Berkelium	7	Actinide series	f	Actinide	[247]	Synthetic	Laboratory synthesis; astrophysical occurrence not a normal abundance class	Nested electron/nuclear channels; tests partial familiarity and radioactive clocks.
98	Cf	Californium	7	Actinide series	f	Actinide	[251]	Synthetic	Laboratory synthesis; astrophysical occurrence not a normal abundance class	Nested electron/nuclear channels; tests partial familiarity and radioactive clocks.
99	Es	Einsteinium	7	Actinide series	f	Actinide	[252]	Synthetic	Laboratory synthesis; astrophysical occurrence not a normal abundance class	Nested electron/nuclear channels; tests partial familiarity and radioactive clocks.
100	Fm	Fermium	7	Actinide series	f	Actinide	[257]	Synthetic	Laboratory synthesis; astrophysical occurrence not a normal abundance class	Nested electron/nuclear channels; tests partial familiarity and radioactive clocks.
101	Md	Mendelevium	7	Actinide series	f	Actinide	[258]	Synthetic	Laboratory synthesis; astrophysical occurrence not a normal abundance class	Nested electron/nuclear channels; tests partial familiarity and radioactive clocks.
102	No	Nobelium	7	Actinide series	f	Actinide	[259]	Synthetic	Laboratory synthesis; astrophysical occurrence not a normal abundance class	Nested electron/nuclear channels; tests partial familiarity and radioactive clocks.
103	Lr	Lawrencium	7	Actinide series	f	Actinide	[266]	Synthetic	Laboratory synthesis; astrophysical occurrence not a normal abundance class	Nested electron/nuclear channels; tests partial familiarity and radioactive clocks.
104	Rf	Rutherfordium	7	Group 4	d	Transition metal	[267]	Synthetic	Laboratory synthesis; astrophysical occurrence not a normal abundance class	Atomic boundary class; relates nuclear identity, electron-shell response and environment-dependent chemistry.
105	Db	Dubnium	7	Group 5	d	Transition metal	[268]	Synthetic	Laboratory synthesis; astrophysical occurrence not a normal abundance class	Atomic boundary class; relates nuclear identity, electron-shell response and environment-dependent chemistry.
106	Sg	Seaborgium	7	Group 6	d	Transition metal	[269]	Synthetic	Laboratory synthesis; astrophysical occurrence not a normal abundance class	Atomic boundary class; relates nuclear identity, electron-shell response and environment-dependent chemistry.
107	Bh	Bohrium	7	Group 7	d	Transition metal	[270]	Synthetic	Laboratory synthesis; astrophysical occurrence not a normal abundance class	Atomic boundary class; relates nuclear identity, electron-shell response and environment-dependent chemistry.
108	Hs	Hassium	7	Group 8	d	Transition metal	[269]	Synthetic	Laboratory synthesis; astrophysical occurrence not a normal abundance class	Atomic boundary class; relates nuclear identity, electron-shell response and environment-dependent chemistry.
109	Mt	Meitnerium	7	Group 9	d	Transition metal	[277]	Synthetic	Laboratory synthesis; astrophysical occurrence not a normal abundance class	Atomic boundary class; relates nuclear identity, electron-shell response and environment-dependent chemistry.

Z	Sym	Element	Period	Group/series	Block	Family	Atomic weight / mass	Isotope/provenance	Origin clock	UM use
110	Ds	Darmstadtium	7	Group 10	d	Transition metal	[281]	Synthetic	Laboratory synthesis; astrophysical occurrence not a normal abundance class	Atomic boundary class; relates nuclear identity, electron-shell response and environment-dependent chemistry.
111	Rg	Roentgenium	7	Group 11	d	Transition metal	[282]	Synthetic	Laboratory synthesis; astrophysical occurrence not a normal abundance class	Atomic boundary class; relates nuclear identity, electron-shell response and environment-dependent chemistry.
112	Cn	Copernicium	7	Group 12	d	Transition metal	[285]	Synthetic	Laboratory synthesis; astrophysical occurrence not a normal abundance class	Atomic boundary class; relates nuclear identity, electron-shell response and environment-dependent chemistry.
113	Nh	Nihonium	7	Group 13	p	Superheavy; chemistry partly predicted	[286]	Synthetic	Laboratory synthesis; astrophysical occurrence not a normal abundance class	Atomic boundary class; relates nuclear identity, electron-shell response and environment-dependent chemistry.
114	Fl	Flerovium	7	Group 14	p	Superheavy; chemistry partly predicted	[290]	Synthetic	Laboratory synthesis; astrophysical occurrence not a normal abundance class	Atomic boundary class; relates nuclear identity, electron-shell response and environment-dependent chemistry.
115	Mc	Moscovium	7	Group 15	p	Superheavy; chemistry partly predicted	[290]	Synthetic	Laboratory synthesis; astrophysical occurrence not a normal abundance class	Atomic boundary class; relates nuclear identity, electron-shell response and environment-dependent chemistry.
116	Lv	Livermorium	7	Group 16	p	Superheavy; chemistry partly predicted	[293]	Synthetic	Laboratory synthesis; astrophysical occurrence not a normal abundance class	Atomic boundary class; relates nuclear identity, electron-shell response and environment-dependent chemistry.
117	Ts	Tennessine	7	Group 17	p	Halogen	[294]	Synthetic	Laboratory synthesis; astrophysical occurrence not a normal abundance class	Valence-boundary class; tests high exchange propensity.
118	Og	Oganesson	7	Group 18	p	Noble gas	[294]	Synthetic	Laboratory synthesis; astrophysical occurrence not a normal abundance class	Closed-shell familiarity class; tests low-reactivity identity retention.

PART V — CROSS-SCALE OBJECT ATLAS

The atlas does not claim that one image, spectrum, or equation contains an entire object. Each entry keeps four clocks, direct records, established inferences, actual Boundaries, channel limits, familiarity classes, retention conditions, and UM reconstruction distinct.

5.1 Planet language classes

PL-001 — Mercury (Rocky planet)

Characteristic radius or scale	2,439.7 km
Mass	3.301e23 kg
Distance from Earth	77–222 million km
Signal age	4.3–12.3 light-min
Formation age	4.5–4.6 Gyr
Material age	Solar-nebula matter; many heavy nuclei were made before the Solar System.
Processing age	Cratering and volcanism; active surface change is slow
Direct records	Images, spectra, thermal maps, timing; for visited planets also fields, gravity and in-situ samples.
Established inference	Bulk mass, radius, atmosphere and interior are joint model inferences; confidence differs by layer.
Actual Boundary at this scale	At image scale: radiating or reflecting atmosphere/surface. At dynamical scale: gravitationally retained planet system.
Dominant channels	Visible/IR/UV/radio light; occultation; spacecraft tracking; gravity; magnetic field; in-situ particles where available.
Atomic-family evidence	Silicate rock, iron-rich core; Na, K, Ca and Mg exosphere
Transmission limit	A photograph alone omits deep composition, isotope history and most interior states.
Familiarity class	Rocky planet with atmosphere, interior and orbital subclasses.
Identity retention	Retained by self-gravity over Gyr while weather, chemistry and impacts exchange information.
Complexity profile	High state diversity; nested atmosphere/interior/orbit; long retention; many partially independent channels.
UM reinterpretation	Treat each instrument as a channel cut. Joint channels shrink the physical blind region; unresolved residuals enter the Familiarity Kernel.
Evidence grade	R/E established; U interpretive
Source URL	Open source

PL-002 — Venus (Rocky planet)

Characteristic radius or scale	6,051.8 km
Mass	4.867e24 kg
Distance from Earth	38–261 million km
Signal age	2.1–14.5 light-min
Formation age	4.5–4.6 Gyr
Material age	Solar-nebula matter; many heavy nuclei were made before the Solar System.
Processing age	Resurfacing and atmospheric chemistry clocks remain model-dependent
Direct records	Images, spectra, thermal maps, timing; for visited planets also fields, gravity and in-situ samples.
Established inference	Bulk mass, radius, atmosphere and interior are joint model inferences; confidence differs by layer.
Actual Boundary at this scale	At image scale: radiating or reflecting atmosphere/surface. At dynamical scale: gravitationally retained planet system.
Dominant channels	Visible/IR/UV/radio light; occultation; spacecraft tracking; gravity; magnetic field; in-situ particles where available.
Atomic-family evidence	CO_2 atmosphere; N, S, O signatures; silicate surface
Transmission limit	A photograph alone omits deep composition, isotope history and most interior states.
Familiarity class	Rocky planet with atmosphere, interior and orbital subclasses.
Identity retention	Retained by self-gravity over Gyr while weather, chemistry and impacts exchange information.
Complexity profile	High state diversity; nested atmosphere/interior/orbit; long retention; many partially independent channels.
UM reinterpretation	Treat each instrument as a channel cut. Joint channels shrink the physical blind region; unresolved residuals enter the Familiarity Kernel.
Evidence grade	R/E established; U interpretive
Source URL	Open source

PL-003 — Earth (Rocky ocean planet)

Characteristic radius or scale	6,371 km mean
Mass	5.9722e24 kg
Distance from Earth	Local observer frame
Signal age	Local / instrument delay
Formation age	4.54 Gyr

Material age	Solar-nebula matter; many heavy nuclei were made before the Solar System.
Processing age	Continuously reprocessed by plate tectonics, water, atmosphere and life
Direct records	Images, spectra, thermal maps, timing; for visited planets also fields, gravity and in-situ samples.
Established inference	Bulk mass, radius, atmosphere and interior are joint model inferences; confidence differs by layer.
Actual Boundary at this scale	At image scale: radiating or reflecting atmosphere/surface. At dynamical scale: gravitationally retained planet system.
Dominant channels	Visible/IR/UV/radio light; occultation; spacecraft tracking; gravity; magnetic field; in-situ particles where available.
Atomic-family evidence	O, Si, Al, Fe, Ca, Na, K, Mg bulk crust; H-C-N-O biosphere
Transmission limit	A photograph alone omits deep composition, isotope history and most interior states.
Familiarity class	Rocky ocean planet with atmosphere, interior and orbital subclasses.
Identity retention	Retained by self-gravity over Gyr while weather, chemistry and impacts exchange information.
Complexity profile	High state diversity; nested atmosphere/interior/orbit; long retention; many partially independent channels.
UM reinterpretation	Treat each instrument as a channel cut. Joint channels shrink the physical blind region; unresolved residuals enter the Familiarity Kernel.
Evidence grade	R/E established; U interpretive
Source URL	Open source

PL-004 — Mars (Rocky planet)

Characteristic radius or scale	3,389.5 km
Mass	6.417e23 kg
Distance from Earth	55–401 million km
Signal age	3.1–22.3 light-min
Formation age	4.5–4.6 Gyr
Material age	Solar-nebula matter; many heavy nuclei were made before the Solar System.
Processing age	Ancient aqueous alteration, volcanism, impacts and present weathering
Direct records	Images, spectra, thermal maps, timing; for visited planets also fields, gravity and in-situ samples.
Established inference	Bulk mass, radius, atmosphere and interior are joint model inferences; confidence differs by layer.
Actual Boundary at this scale	At image scale: radiating or reflecting atmosphere/surface. At dynamical scale: gravitationally retained planet system.

Dominant channels	Visible/IR/UV/radio light; occultation; spacecraft tracking; gravity; magnetic field; in-situ particles where available.
Atomic-family evidence	Fe oxides, silicates, CO2 atmosphere, H2O ice
Transmission limit	A photograph alone omits deep composition, isotope history and most interior states.
Familiarity class	Rocky planet with atmosphere, interior and orbital subclasses.
Identity retention	Retained by self-gravity over Gyr while weather, chemistry and impacts exchange information.
Complexity profile	High state diversity; nested atmosphere/interior/orbit; long retention; many partially independent channels.
UM reinterpretation	Treat each instrument as a channel cut. Joint channels shrink the physical blind region; unresolved residuals enter the Familiarity Kernel.
Evidence grade	R/E established; U interpretive
Source URL	Open source

PL-005 — Jupiter (Gas giant)

Characteristic radius or scale	69,911 km mean
Mass	1.898e27 kg
Distance from Earth	588–968 million km
Signal age	32.7–53.8 light-min
Formation age	4.5–4.6 Gyr
Material age	Solar-nebula matter; many heavy nuclei were made before the Solar System.
Processing age	Atmospheric circulation is current; interior differentiation is inferred
Direct records	Images, spectra, thermal maps, timing; for visited planets also fields, gravity and in-situ samples.
Established inference	Bulk mass, radius, atmosphere and interior are joint model inferences; confidence differs by layer.
Actual Boundary at this scale	At image scale: radiating or reflecting atmosphere/surface. At dynamical scale: gravitationally retained planet system.
Dominant channels	Visible/IR/UV/radio light; occultation; spacecraft tracking; gravity; magnetic field; in-situ particles where available.
Atomic-family evidence	H and He dominate; CH4, NH3, H2O and trace heavy elements
Transmission limit	A photograph alone omits deep composition, isotope history and most interior states.
Familiarity class	Gas giant with atmosphere, interior and orbital subclasses.
Identity retention	Retained by self-gravity over Gyr while weather, chemistry and impacts exchange information.

Complexity profile	High state diversity; nested atmosphere/interior/orbit; long retention; many partially independent channels.
UM reinterpretation	Treat each instrument as a channel cut. Joint channels shrink the physical blind region; unresolved residuals enter the Familiarity Kernel.
Evidence grade	R/E established; U interpretive
Source URL	Open source

PL-006 — Saturn (Gas giant)

Characteristic radius or scale	58,232 km mean
Mass	5.683e26 kg
Distance from Earth	1.2–1.67 billion km
Signal age	67–93 light-min
Formation age	4.5–4.6 Gyr
Material age	Solar-nebula matter; many heavy nuclei were made before the Solar System.
Processing age	Atmosphere, rings and interior each have different processing clocks
Direct records	Images, spectra, thermal maps, timing; for visited planets also fields, gravity and in-situ samples.
Established inference	Bulk mass, radius, atmosphere and interior are joint model inferences; confidence differs by layer.
Actual Boundary at this scale	At image scale: radiating or reflecting atmosphere/surface. At dynamical scale: gravitationally retained planet system.
Dominant channels	Visible/IR/UV/radio light; occultation; spacecraft tracking; gravity; magnetic field; in-situ particles where available.
Atomic-family evidence	H and He dominate; CH_4, NH_3, H_2O; ring water ice
Transmission limit	A photograph alone omits deep composition, isotope history and most interior states.
Familiarity class	Gas giant with atmosphere, interior and orbital subclasses.
Identity retention	Retained by self-gravity over Gyr while weather, chemistry and impacts exchange information.
Complexity profile	High state diversity; nested atmosphere/interior/orbit; long retention; many partially independent channels.
UM reinterpretation	Treat each instrument as a channel cut. Joint channels shrink the physical blind region; unresolved residuals enter the Familiarity Kernel.
Evidence grade	R/E established; U interpretive
Source URL	Open source

PL-007 — Uranus (Ice giant)

Characteristic radius or scale	25,362 km mean
Mass	8.681e25 kg
Distance from Earth	2.6–3.2 billion km
Signal age	145–178 light-min
Formation age	4.5–4.6 Gyr
Material age	Solar-nebula matter; many heavy nuclei were made before the Solar System.
Processing age	Interior thermal history and atmospheric circulation remain incomplete
Direct records	Images, spectra, thermal maps, timing; for visited planets also fields, gravity and in-situ samples.
Established inference	Bulk mass, radius, atmosphere and interior are joint model inferences; confidence differs by layer.
Actual Boundary at this scale	At image scale: radiating or reflecting atmosphere/surface. At dynamical scale: gravitationally retained planet system.
Dominant channels	Visible/IR/UV/radio light; occultation; spacecraft tracking; gravity; magnetic field; in-situ particles where available.
Atomic-family evidence	H, He, CH4 atmosphere; water/ammonia/methane-rich interior models
Transmission limit	A photograph alone omits deep composition, isotope history and most interior states.
Familiarity class	Ice giant with atmosphere, interior and orbital subclasses.
Identity retention	Retained by self-gravity over Gyr while weather, chemistry and impacts exchange information.
Complexity profile	High state diversity; nested atmosphere/interior/orbit; long retention; many partially independent channels.
UM reinterpretation	Treat each instrument as a channel cut. Joint channels shrink the physical blind region; unresolved residuals enter the Familiarity Kernel.
Evidence grade	R/E established; U interpretive
Source URL	Open source

PL-008 — Neptune (Ice giant)

Characteristic radius or scale	24,622 km mean
Mass	1.024e26 kg
Distance from Earth	4.3–4.7 billion km
Signal age	239–261 light-min
Formation age	4.5–4.6 Gyr

Material age	Solar-nebula matter; many heavy nuclei were made before the Solar System.
Processing age	Active weather over a long interior cooling history
Direct records	Images, spectra, thermal maps, timing; for visited planets also fields, gravity and in-situ samples.
Established inference	Bulk mass, radius, atmosphere and interior are joint model inferences; confidence differs by layer.
Actual Boundary at this scale	At image scale: radiating or reflecting atmosphere/surface. At dynamical scale: gravitationally retained planet system.
Dominant channels	Visible/IR/UV/radio light; occultation; spacecraft tracking; gravity; magnetic field; in-situ particles where available.
Atomic-family evidence	H, He, CH4 atmosphere; volatile-rich interior models
Transmission limit	A photograph alone omits deep composition, isotope history and most interior states.
Familiarity class	Ice giant with atmosphere, interior and orbital subclasses.
Identity retention	Retained by self-gravity over Gyr while weather, chemistry and impacts exchange information.
Complexity profile	High state diversity; nested atmosphere/interior/orbit; long retention; many partially independent channels.
UM reinterpretation	Treat each instrument as a channel cut. Joint channels shrink the physical blind region; unresolved residuals enter the Familiarity Kernel.
Evidence grade	R/E established; U interpretive
Source URL	Open source

5.2 Stellar language classes

ST-001 — Sun (G-type main-sequence star)

Characteristic radius or scale	695,700 km radius
Mass	1.989e30 kg
Distance from Earth	1 AU
Signal age	8.3 light-min
Formation age	4.57 Gyr
Material age	Constituent nuclei span primordial material and earlier stellar generations; the star also makes new nuclei.
Processing age	Core fusion, convection, rotation and magnetic-cycle clocks
Direct records	Spectra, brightness and timing records; resolved images only for selected stars; neutrinos available for the Sun.

Established inference	Temperature, radius, mass, composition and evolutionary state are joint inferences with class-dependent uncertainty.
Actual Boundary at this scale	Photosphere for ordinary light; gravitationally retained plasma for dynamics; compact-matter surface or emission region for remnants.
Dominant channels	Multi-band photons, Doppler shifts, astrometry, asteroseismology, binaries; solar neutrinos; pulsar timing for neutron stars.
Atomic-family evidence	H/He spectrum plus trace-element absorption lines
Transmission limit	One spectrum or image omits the complete core state, three-dimensional history and most internal trajectories.
Familiarity class	G-type main-sequence star plus spectral, variability and evolutionary subclasses.
Identity retention	Self-gravity retains the star while fusion, radiation, winds and rotation exchange energy and matter.
Complexity profile	Very high state diversity and nesting; multiple clocks; channel coverage ranges from rich for the Sun to sparse for distant stars.
UM reinterpretation	The star is a hierarchy of causal boundaries, not one state variable. UM should compile channel-consistent identities before proposing extra dynamics.
Evidence grade	R/E established; U interpretive
Source URL	Open source

ST-002 — Proxima Centauri (M-type red dwarf)

Characteristic radius or scale	about 0.154 solar radii
Mass	about 0.122 solar masses
Distance from Earth	4.2465 light-years
Signal age	4.2465 years
Formation age	Model-dependent; several Gyr
Material age	Constituent nuclei span primordial material and earlier stellar generations; the star also makes new nuclei.
Processing age	Main-sequence burning with flare and magnetic clocks
Direct records	Spectra, brightness and timing records; resolved images only for selected stars; neutrinos available for the Sun.
Established inference	Temperature, radius, mass, composition and evolutionary state are joint inferences with class-dependent uncertainty.
Actual Boundary at this scale	Photosphere for ordinary light; gravitationally retained plasma for dynamics; compact-matter surface or emission region for remnants.
Dominant channels	Multi-band photons, Doppler shifts, astrometry, asteroseismology, binaries; solar neutrinos; pulsar timing for neutron stars.
Atomic-family evidence	Cool-star atomic and molecular spectra

Transmission limit	One spectrum or image omits the complete core state, three-dimensional history and most internal trajectories.
Familiarity class	M-type red dwarf plus spectral, variability and evolutionary subclasses.
Identity retention	Self-gravity retains the star while fusion, radiation, winds and rotation exchange energy and matter.
Complexity profile	Very high state diversity and nesting; multiple clocks; channel coverage ranges from rich for the Sun to sparse for distant stars.
UM reinterpretation	The star is a hierarchy of causal boundaries, not one state variable. UM should compile channel-consistent identities before proposing extra dynamics.
Evidence grade	R/E established; U interpretive
Source URL	Open source

ST-003 — Sirius A (A-type main-sequence star)

Characteristic radius or scale	about 1.7 solar radii
Mass	about 2.1 solar masses
Distance from Earth	8.6 light-years
Signal age	8.6 years
Formation age	System age about 0.2–0.3 Gyr
Material age	Constituent nuclei span primordial material and earlier stellar generations; the star also makes new nuclei.
Processing age	Rapid main-sequence evolution; binary orbital clock
Direct records	Spectra, brightness and timing records; resolved images only for selected stars; neutrinos available for the Sun.
Established inference	Temperature, radius, mass, composition and evolutionary state are joint inferences with class-dependent uncertainty.
Actual Boundary at this scale	Photosphere for ordinary light; gravitationally retained plasma for dynamics; compact-matter surface or emission region for remnants.
Dominant channels	Multi-band photons, Doppler shifts, astrometry, asteroseismology, binaries; solar neutrinos; pulsar timing for neutron stars.
Atomic-family evidence	Hot photospheric absorption spectrum
Transmission limit	One spectrum or image omits the complete core state, three-dimensional history and most internal trajectories.
Familiarity class	A-type main-sequence star plus spectral, variability and evolutionary subclasses.
Identity retention	Self-gravity retains the star while fusion, radiation, winds and rotation exchange energy and matter.
Complexity profile	Very high state diversity and nesting; multiple clocks; channel coverage ranges from rich for the Sun to sparse for distant stars.

UM reinterpretation	The star is a hierarchy of causal boundaries, not one state variable. UM should compile channel-consistent identities before proposing extra dynamics.
Evidence grade	R/E established; U interpretive
Source URL	Open source

ST-004 — Betelgeuse (Red supergiant)

Characteristic radius or scale	about 700 solar radii
Mass	about 15 solar masses
Distance from Earth	about 700 light-years
Signal age	about 700 years
Formation age	about 10 million years
Material age	Constituent nuclei span primordial material and earlier stellar generations; the star also makes new nuclei.
Processing age	Pulsation, convection, mass loss and late fusion stages
Direct records	Spectra, brightness and timing records; resolved images only for selected stars; neutrinos available for the Sun.
Established inference	Temperature, radius, mass, composition and evolutionary state are joint inferences with class-dependent uncertainty.
Actual Boundary at this scale	Photosphere for ordinary light; gravitationally retained plasma for dynamics; compact-matter surface or emission region for remnants.
Dominant channels	Multi-band photons, Doppler shifts, astrometry, asteroseismology, binaries; solar neutrinos; pulsar timing for neutron stars.
Atomic-family evidence	Cool atmosphere with atomic and molecular signatures
Transmission limit	One spectrum or image omits the complete core state, three-dimensional history and most internal trajectories.
Familiarity class	Red supergiant plus spectral, variability and evolutionary subclasses.
Identity retention	Self-gravity retains the star while fusion, radiation, winds and rotation exchange energy and matter.
Complexity profile	Very high state diversity and nesting; multiple clocks; channel coverage ranges from rich for the Sun to sparse for distant stars.
UM reinterpretation	The star is a hierarchy of causal boundaries, not one state variable. UM should compile channel-consistent identities before proposing extra dynamics.
Evidence grade	R/E established; U interpretive
Source URL	Open source

ST-005 — Sirius B (White dwarf)

Characteristic radius or scale	about 0.0084 solar radii
Mass	about 1.02 solar masses
Distance from Earth	8.6 light-years
Signal age	8.6 years
Formation age	System age about 0.2–0.3 Gyr
Material age	Constituent nuclei span primordial material and earlier stellar generations; the star also makes new nuclei.
Processing age	White-dwarf cooling clock after progenitor evolution
Direct records	Spectra, brightness and timing records; resolved images only for selected stars; neutrinos available for the Sun.
Established inference	Temperature, radius, mass, composition and evolutionary state are joint inferences with class-dependent uncertainty.
Actual Boundary at this scale	Photosphere for ordinary light; gravitationally retained plasma for dynamics; compact-matter surface or emission region for remnants.
Dominant channels	Multi-band photons, Doppler shifts, astrometry, asteroseismology, binaries; solar neutrinos; pulsar timing for neutron stars.
Atomic-family evidence	Hydrogen-dominated atmosphere over degenerate C/O interior model
Transmission limit	One spectrum or image omits the complete core state, three-dimensional history and most internal trajectories.
Familiarity class	White dwarf plus spectral, variability and evolutionary subclasses.
Identity retention	Self-gravity retains the star while fusion, radiation, winds and rotation exchange energy and matter.
Complexity profile	Very high state diversity and nesting; multiple clocks; channel coverage ranges from rich for the Sun to sparse for distant stars.
UM reinterpretation	The star is a hierarchy of causal boundaries, not one state variable. UM should compile channel-consistent identities before proposing extra dynamics.
Evidence grade	R/E established; U interpretive
Source URL	Open source

ST-006 — PSR J0740+6620 (Millisecond pulsar / neutron star)

Characteristic radius or scale	roughly 12 km; model-dependent
Mass	about 2.1 solar masses
Distance from Earth	roughly 3,000 light-years
Signal age	roughly 3,000 years

Formation age	Spin-down and binary history are model-dependent
Material age	Constituent nuclei span primordial material and earlier stellar generations; the star also makes new nuclei.
Processing age	Millisecond rotation, accretion history and cooling clocks
Direct records	Spectra, brightness and timing records; resolved images only for selected stars; neutrinos available for the Sun.
Established inference	Temperature, radius, mass, composition and evolutionary state are joint inferences with class-dependent uncertainty.
Actual Boundary at this scale	Photosphere for ordinary light; gravitationally retained plasma for dynamics; compact-matter surface or emission region for remnants.
Dominant channels	Multi-band photons, Doppler shifts, astrometry, asteroseismology, binaries; solar neutrinos; pulsar timing for neutron stars.
Atomic-family evidence	Nuclear-matter composition inferred, not directly sampled
Transmission limit	One spectrum or image omits the complete core state, three-dimensional history and most internal trajectories.
Familiarity class	Millisecond pulsar / neutron star plus spectral, variability and evolutionary subclasses.
Identity retention	Self-gravity retains the star while fusion, radiation, winds and rotation exchange energy and matter.
Complexity profile	Very high state diversity and nesting; multiple clocks; channel coverage ranges from rich for the Sun to sparse for distant stars.
UM reinterpretation	The star is a hierarchy of causal boundaries, not one state variable. UM should compile channel-consistent identities before proposing extra dynamics.
Evidence grade	R/E established; U interpretive
Source URL	Open source

5.3 Galaxy language classes

GA-001 — Milky Way (Barred spiral galaxy)

Characteristic radius or scale	>100,000 light-year stellar disk
Mass	about 10^12 solar masses including halo; model-dependent
Distance from Earth	Observer is inside it
Signal age	0 to >100,000 years across the disk
Formation age	Oldest populations >13 Gyr; assembly extended
Material age	Gas and stellar nuclei span primordial matter and many generations of stellar nucleosynthesis.
Processing age	Ongoing star formation, enrichment, orbital mixing and mergers

Direct records	Images, spectra, redshifts, stellar motions, gas velocities, lensing, radio and X-ray maps.
Established inference	Stellar mass, dark-matter distribution, star-formation history and merger history are model-dependent joint inferences.
Actual Boundary at this scale	No single surface: stellar disk, gas, halo, group or cluster are nested causal retention domains.
Dominant channels	UV/optical/IR/radio/X-ray light; resolved motions when near; lensing and satellite dynamics for gravitating mass.
Atomic-family evidence	Resolved stellar spectra, gas lines, dust bands and abundance gradients
Transmission limit	Integrated light erases most individual stellar states; projected images erase depth; histories are many-to-one.
Familiarity class	Barred spiral galaxy with stellar-population, gas, halo, activity and environment subclasses.
Identity retention	Gravity retains the ensemble over many orbital periods while stars, gas, radiation and satellites exchange matter and information.
Complexity profile	Extreme population diversity, deep nesting, long retention and heavy model dependence; many channels still leave degeneracies.
UM reinterpretation	A galaxy is a ledger of interacting familiar classes. UM should preserve separate disk, halo, gas, black-hole and environmental boundaries.
Evidence grade	R/E established; U interpretive
Source URL	Open source

GA-002 — Andromeda / M31 (Barred spiral galaxy)

Characteristic radius or scale	roughly 200,000 light-year stellar extent
Mass	order 10^12 solar masses; model-dependent
Distance from Earth	2.5 million light-years
Signal age	2.5 million years
Formation age	Old populations plus prolonged assembly
Material age	Gas and stellar nuclei span primordial matter and many generations of stellar nucleosynthesis.
Processing age	Star formation, accretion and merger history
Direct records	Images, spectra, redshifts, stellar motions, gas velocities, lensing, radio and X-ray maps.
Established inference	Stellar mass, dark-matter distribution, star-formation history and merger history are model-dependent joint inferences.
Actual Boundary at this scale	No single surface: stellar disk, gas, halo, group or cluster are nested causal retention domains.
Dominant channels	UV/optical/IR/radio/X-ray light; resolved motions when near; lensing and satellite dynamics for gravitating mass.
Atomic-family evidence	Resolved stars, gas and dust; metallicity and population gradients

Transmission limit	Integrated light erases most individual stellar states; projected images erase depth; histories are many-to-one.
Familiarity class	Barred spiral galaxy with stellar-population, gas, halo, activity and environment subclasses.
Identity retention	Gravity retains the ensemble over many orbital periods while stars, gas, radiation and satellites exchange matter and information.
Complexity profile	Extreme population diversity, deep nesting, long retention and heavy model dependence; many channels still leave degeneracies.
UM reinterpretation	A galaxy is a ledger of interacting familiar classes. UM should preserve separate disk, halo, gas, black-hole and environmental boundaries.
Evidence grade	R/E established; U interpretive
Source URL	Open source

GA-003 — M87 (Giant elliptical galaxy)

Characteristic radius or scale	tens to hundreds of kiloparsecs by component
Mass	Massive galaxy plus cluster-scale halo; model-dependent
Distance from Earth	54 million light-years
Signal age	54 million years
Formation age	Old stellar population with merger-built assembly
Material age	Gas and stellar nuclei span primordial matter and many generations of stellar nucleosynthesis.
Processing age	Jet, hot gas, black-hole accretion and cluster interaction clocks
Direct records	Images, spectra, redshifts, stellar motions, gas velocities, lensing, radio and X-ray maps.
Established inference	Stellar mass, dark-matter distribution, star-formation history and merger history are model-dependent joint inferences.
Actual Boundary at this scale	No single surface: stellar disk, gas, halo, group or cluster are nested causal retention domains.
Dominant channels	UV/optical/IR/radio/X-ray light; resolved motions when near; lensing and satellite dynamics for gravitating mass.
Atomic-family evidence	Old stellar spectra, X-ray gas and relativistic jet plasma
Transmission limit	Integrated light erases most individual stellar states; projected images erase depth; histories are many-to-one.
Familiarity class	Giant elliptical galaxy with stellar-population, gas, halo, activity and environment subclasses.
Identity retention	Gravity retains the ensemble over many orbital periods while stars, gas, radiation and satellites exchange matter and information.
Complexity profile	Extreme population diversity, deep nesting, long retention and heavy model dependence; many channels still leave degeneracies.

UM reinterpretation	A galaxy is a ledger of interacting familiar classes. UM should preserve separate disk, halo, gas, black-hole and environmental boundaries.
Evidence grade	R/E established; U interpretive
Source URL	Open source

GA-004 — Small Magellanic Cloud (Dwarf irregular galaxy)

Characteristic radius or scale	roughly 7,000 light-years
Mass	few billion solar masses; model-dependent
Distance from Earth	roughly 200,000 light-years
Signal age	roughly 200,000 years
Formation age	Extended star-formation and interaction history
Material age	Gas and stellar nuclei span primordial matter and many generations of stellar nucleosynthesis.
Processing age	Current star formation and tidal interaction with the Milky Way
Direct records	Images, spectra, redshifts, stellar motions, gas velocities, lensing, radio and X-ray maps.
Established inference	Stellar mass, dark-matter distribution, star-formation history and merger history are model-dependent joint inferences.
Actual Boundary at this scale	No single surface: stellar disk, gas, halo, group or cluster are nested causal retention domains.
Dominant channels	UV/optical/IR/radio/X-ray light; resolved motions when near; lensing and satellite dynamics for gravitating mass.
Atomic-family evidence	Low-metallicity stars, ionized gas and dust
Transmission limit	Integrated light erases most individual stellar states; projected images erase depth; histories are many-to-one.
Familiarity class	Dwarf irregular galaxy with stellar-population, gas, halo, activity and environment subclasses.
Identity retention	Gravity retains the ensemble over many orbital periods while stars, gas, radiation and satellites exchange matter and information.
Complexity profile	Extreme population diversity, deep nesting, long retention and heavy model dependence; many channels still leave degeneracies.
UM reinterpretation	A galaxy is a ledger of interacting familiar classes. UM should preserve separate disk, halo, gas, black-hole and environmental boundaries.
Evidence grade	R/E established; U interpretive
Source URL	Open source

GA-005 — M82 (Starburst galaxy)

Characteristic radius or scale	roughly 37,000 light-years
Mass	tens of billions of solar masses; model-dependent
Distance from Earth	about 12 million light-years
Signal age	about 12 million years
Formation age	Older galaxy with a recent interaction-driven starburst
Material age	Gas and stellar nuclei span primordial matter and many generations of stellar nucleosynthesis.
Processing age	Rapid star formation, winds, supernovae and feedback
Direct records	Images, spectra, redshifts, stellar motions, gas velocities, lensing, radio and X-ray maps.
Established inference	Stellar mass, dark-matter distribution, star-formation history and merger history are model-dependent joint inferences.
Actual Boundary at this scale	No single surface: stellar disk, gas, halo, group or cluster are nested causal retention domains.
Dominant channels	UV/optical/IR/radio/X-ray light; resolved motions when near; lensing and satellite dynamics for gravitating mass.
Atomic-family evidence	Emission lines, hot gas, dust and supernova-enriched material
Transmission limit	Integrated light erases most individual stellar states; projected images erase depth; histories are many-to-one.
Familiarity class	Starburst galaxy with stellar-population, gas, halo, activity and environment subclasses.
Identity retention	Gravity retains the ensemble over many orbital periods while stars, gas, radiation and satellites exchange matter and information.
Complexity profile	Extreme population diversity, deep nesting, long retention and heavy model dependence; many channels still leave degeneracies.
UM reinterpretation	A galaxy is a ledger of interacting familiar classes. UM should preserve separate disk, halo, gas, black-hole and environmental boundaries.
Evidence grade	R/E established; U interpretive
Source URL	Open source

5.4 Black-hole language classes

BH-001 — Sagittarius A* (Galactic-centre supermassive black hole)

Characteristic radius or scale	r_s about 12 million km for 4.0e6 solar masses
Mass	4.0e6 solar masses from EHT calibration; other orbital estimates are more precise
Distance from Earth	about 26,000 light-years

Signal age	about 26,000 years
Formation age	Formation history not directly dated
Material age	Progenitor or accreted matter has older nuclear clocks; the black-hole record does not preserve all of them.
Processing age	Minute-to-hour accretion variability; stellar orbits span years to decades
Direct records	Emission from surrounding matter, stellar or gas dynamics, interferometric visibilities, X-rays or gravitational-wave strain.
Established inference	Mass, spin, compactness and horizon consistency are inferred through general-relativistic models and environmental models.
Actual Boundary at this scale	Event horizon is a theoretical causal boundary consistent with data; the measured bright ring or accretion flow is not the horizon itself.
Dominant channels	Orbital dynamics; radio interferometry; optical/IR/X-ray plasma; gravitational waves, depending on object.
Atomic-family evidence	Ionized accretion plasma; stellar spectra and maser/astrometric references
Transmission limit	External records do not encode a recoverable microscopic interior history; plasma emission also has radiative-transfer degeneracies.
Familiarity class	Galactic-centre supermassive black hole with mass, spin, accretion, environment and waveform subclasses.
Identity retention	Externally characterized by long-lived mass/spin/charge parameters while accretion and environment remain dynamic.
Complexity profile	Few external parameters but highly layered inference: detector, environment, relativistic transfer and population history.
UM reinterpretation	This is the strongest warning against confusing simple outward descriptors with simple underlying history. UM can index the causal cut but cannot infer erased microhistory.
Evidence grade	R/E established; U interpretive
Source URL	Open source

BH-002 — M87 (Supermassive black hole)*

Characteristic radius or scale	r_s about 19 billion km for 6.5e9 solar masses
Mass	6.5e9 solar masses with statistical and systematic uncertainty
Distance from Earth	54 million light-years
Signal age	54 million years
Formation age	Formation and growth history inferred from galaxy evolution
Material age	Progenitor or accreted matter has older nuclear clocks; the black-hole record does not preserve all of them.
Processing age	Horizon-scale emission, accretion and jet clocks

Direct records	Emission from surrounding matter, stellar or gas dynamics, interferometric visibilities, X-rays or gravitational-wave strain.
Established inference	Mass, spin, compactness and horizon consistency are inferred through general-relativistic models and environmental models.
Actual Boundary at this scale	Event horizon is a theoretical causal boundary consistent with data; the measured bright ring or accretion flow is not the horizon itself.
Dominant channels	Orbital dynamics; radio interferometry; optical/IR/X-ray plasma; gravitational waves, depending on object.
Atomic-family evidence	Synchrotron-emitting plasma and multi-band jet material
Transmission limit	External records do not encode a recoverable microscopic interior history; plasma emission also has radiative-transfer degeneracies.
Familiarity class	Supermassive black hole with mass, spin, accretion, environment and waveform subclasses.
Identity retention	Externally characterized by long-lived mass/spin/charge parameters while accretion and environment remain dynamic.
Complexity profile	Few external parameters but highly layered inference: detector, environment, relativistic transfer and population history.
UM reinterpretation	This is the strongest warning against confusing simple outward descriptors with simple underlying history. UM can index the causal cut but cannot infer erased microhistory.
Evidence grade	R/E established; U interpretive
Source URL	Open source

BH-003 — Cygnus X-1 (Stellar-mass black-hole binary)

Characteristic radius or scale	r_s about 63 km for 21.2 solar masses
Mass	21.2 ± 2.2 solar masses
Distance from Earth	2.22 +0.18/-0.17 kpc
Signal age	about 7,200 years
Formation age	Massive-star remnant; exact formation epoch model-dependent
Material age	Progenitor or accreted matter has older nuclear clocks; the black-hole record does not preserve all of them.
Processing age	Binary orbit, accretion-state and X-ray variability clocks
Direct records	Emission from surrounding matter, stellar or gas dynamics, interferometric visibilities, X-rays or gravitational-wave strain.
Established inference	Mass, spin, compactness and horizon consistency are inferred through general-relativistic models and environmental models.
Actual Boundary at this scale	Event horizon is a theoretical causal boundary consistent with data; the measured bright ring or accretion flow is not the horizon itself.
Dominant channels	Orbital dynamics; radio interferometry; optical/IR/X-ray plasma; gravitational waves, depending on object.

Atomic-family evidence	Companion-star spectrum and accretion-disk atomic lines
Transmission limit	External records do not encode a recoverable microscopic interior history; plasma emission also has radiative-transfer degeneracies.
Familiarity class	Stellar-mass black-hole binary with mass, spin, accretion, environment and waveform subclasses.
Identity retention	Externally characterized by long-lived mass/spin/charge parameters while accretion and environment remain dynamic.
Complexity profile	Few external parameters but highly layered inference: detector, environment, relativistic transfer and population history.
UM reinterpretation	This is the strongest warning against confusing simple outward descriptors with simple underlying history. UM can index the causal cut but cannot infer erased microhistory.
Evidence grade	R/E established; U interpretive
Source URL	Open source

BH-004 — GW150914 remnant (Binary-merger black-hole remnant)

Characteristic radius or scale	r_s about 180 km for roughly 62 solar masses
Mass	roughly 62 solar masses after radiated gravitational-wave energy
Distance from Earth	luminosity distance about 410 Mpc in discovery analysis
Signal age	about 1.3 billion years
Formation age	Remnant formed at the observed merger epoch
Material age	Progenitor or accreted matter has older nuclear clocks; the black-hole record does not preserve all of them.
Processing age	Sub-second merger and ringdown clock
Direct records	Emission from surrounding matter, stellar or gas dynamics, interferometric visibilities, X-rays or gravitational-wave strain.
Established inference	Mass, spin, compactness and horizon consistency are inferred through general-relativistic models and environmental models.
Actual Boundary at this scale	Event horizon is a theoretical causal boundary consistent with data; the measured bright ring or accretion flow is not the horizon itself.
Dominant channels	Orbital dynamics; radio interferometry; optical/IR/X-ray plasma; gravitational waves, depending on object.
Atomic-family evidence	No atomic composition channel in the gravitational-wave record
Transmission limit	External records do not encode a recoverable microscopic interior history; plasma emission also has radiative-transfer degeneracies.
Familiarity class	Binary-merger black-hole remnant with mass, spin, accretion, environment and waveform subclasses.
Identity retention	Externally characterized by long-lived mass/spin/charge parameters while accretion and environment remain dynamic.

Complexity profile	Few external parameters but highly layered inference: detector, environment, relativistic transfer and population history.
UM reinterpretation	This is the strongest warning against confusing simple outward descriptors with simple underlying history. UM can index the causal cut but cannot infer erased microhistory.
Evidence grade	R/E established; U interpretive
Source URL	Open source

5.5 Persons and technology as a local causal language

intent → action → device → encoded network record → computation → returned record → interpretation

A human–AI conversation exists through a local causal chain and becomes meaningful to receivers with compatible language and records. The universe does not require a global state variable stating that the conversation occurred. This example demonstrates that real local events can be omitted from a coarse cosmological description without becoming unreal.

5.6 Compiler visualization

U	Didactic visualization, not a physical mechanism.

A sentence may be visualized as a record that is tokenized, assigned relations, evolved through an internal phase or rotation, and compiled into a small world of stable objects and links. The visualization is useful because it shows how a compact record can generate a relational model. It must not be used as evidence that literal words or sentences physically spin. The physical content begins only after a mechanism and record map are specified.

PART VI — OBSERVABLE MATHEMATICS

Every equation begins with records and ends with a bounded inference. The equation cards below preserve inputs, outputs, validity Boundaries, physical channels, and UM interpretation. A formula is not a universal decoder; its domain assumptions are part of the observable.

OM-001 — Atomic transition

$$\text{Delta E} = \text{h nu} = \text{h c / lambda}$$

Direct inputs (R)	Calibrated spectral line wavelength or frequency
Inferred output (E)	Energy-level difference and element/ion candidates
Validity boundary	Line identification, calibration, broadening and radiative environment
Physical channel	Spectroscopy
UM mapping	Defines an atomic familiarity token; it does not alone fix abundance or history.
Grade	E
Source	Open source

OM-002 — Spectral redshift

$$\text{z} = \text{(lambda\_obs - lambda\_rest) / lambda\_rest}$$

Direct inputs (R)	Observed and laboratory wavelengths
Inferred output (E)	Relative shift; velocity only in an appropriate regime
Validity boundary	Separate Doppler, gravitational and cosmological contributions
Physical channel	Spectroscopy
UM mapping	Maps a preserved wavelength relation into a motion/gravity/expansion class.
Grade	E
Source	Open source

OM-003 — Inverse-square flux

$$F = L / (4 \pi d^2)$$

Direct inputs (R)	Flux plus independently constrained distance or luminosity
Inferred output (E)	Luminosity or distance
Validity boundary	Isotropic propagation, geometry and attenuation assumptions
Physical channel	Photometry
UM mapping	Distance creates a signal gradient, but attenuation and beaming are separate boundaries.
Grade	E
Source	Open source

OM-004 — Thermal luminosity

$$L = 4 \pi R^2 \sigma T_{eff}^4$$

Direct inputs (R)	Bolometric flux, distance and spectral temperature
Inferred output (E)	Effective radius or luminosity
Validity boundary	Blackbody/effective-temperature approximation
Physical channel	Photometry + spectrum
UM mapping	Joint channels convert a light record into an effective emitting-boundary class.
Grade	E
Source	Open source

OM-005 — Wien displacement

$$\lambda_{max} T = b$$

Direct inputs (R)	Peak of an approximately thermal spectrum
Inferred output (E)	Colour/effective temperature

Validity boundary	Requires a near-thermal continuum and known spectral convention
Physical channel	Spectrum
UM mapping	A compact temperature token from a continuum shape; not a complete microstate.
Grade	E
Source	Open source

OM-006 — Line-of-sight Doppler

Delta lambda / lambda approximately v_r / c

Direct inputs (R)	Line shift relative to a rest standard
Inferred output (E)	Radial velocity in the low-speed limit
Validity boundary	Relativistic, gravitational and convective shifts may matter
Physical channel	Spectroscopy
UM mapping	Motion familiarity class carried by relative line displacement.
Grade	E
Source	Open source

OM-007 — Orbital mass

M approximately 4 pi^2 a^3 / (G P^2)

Direct inputs (R)	Orbital scale and period
Inferred output (E)	Enclosed or system mass
Validity boundary	Two-body or modelled multi-body dynamics; inclination and distance
Physical channel	Astrometry + timing
UM mapping	A dynamical mass class; composition remains in the Transmission Kernel of this channel.
Grade	E
Source	Open source

OM-008 — Transit depth

$$\delta \approx (R_p / R_{star})^2$$

Direct inputs (R)	Relative light-curve decrement
Inferred output (E)	Radius ratio
Validity boundary	Geometry, limb darkening, spots and contamination
Physical channel	Time-series photometry
UM mapping	A repeated boundary-crossing token rather than a direct image of the planet.
Grade	E
Source	Open source

OM-009 — Radiative transfer

$$dI_\nu/ds = -\alpha_\nu I_\nu + j_\nu$$

Direct inputs (R)	Intensity as a function of wavelength and path model
Inferred output (E)	Opacity, source function, temperature or abundance constraints
Validity boundary	Geometry, local thermodynamic equilibrium and line model choices
Physical channel	Spectroscopy
UM mapping	Explains why spectra are channel-weighted reconstructions, not transparent inventories.
Grade	E
Source	Open source

OM-010 — Hydrostatic balance

$$dP/dr = -G\, m(r)\, \rho(r) / r^2$$

Direct inputs (R)	Boundary conditions and equation of state

Inferred output (E)	Interior pressure-density profile
Validity boundary	Approximate equilibrium and assumed equation of state
Physical channel	Gravity + thermodynamics
UM mapping	An interior identity model built from boundary data; not directly observed layer by layer.
Grade	E
Source	Open source

OM-011 — Virial scale

M approximately R v^2 / G

Direct inputs (R)	Size and velocity dispersion
Inferred output (E)	Dynamical mass scale
Validity boundary	Equilibrium, geometry and tracer-bias assumptions
Physical channel	Imaging + spectroscopy
UM mapping	Makes gravitationally familiar matter visible through motion even when photon-dark.
Grade	E
Source	Open source

OM-012 — Einstein angle

theta_E^2 = (4GM/c^2)(D_ls/(D_l D_s))

Direct inputs (R)	Image geometry and source/lens distances
Inferred output (E)	Projected lens mass scale
Validity boundary	Lens model, line-of-sight structure and geometry
Physical channel	Gravitational lensing
UM mapping	A joint-channel route that can shrink the photon Transmission Kernel for mass.
Grade	E

Source	Open source

OM-013 — Schwarzschild scale

$$r\_s = 2 G M / c^2$$

Direct inputs (R)	Mass inferred from dynamics or waveform
Inferred output (E)	Characteristic non-spinning horizon scale
Validity boundary	General relativity; spin and environment require richer models
Physical channel	Dynamics / GW / EHT
UM mapping	Defines a causal-scale token, not an image of an interior.
Grade	E
Source	Open source

OM-014 — Eddington luminosity

$$L\_Edd = 4 pi G M m\_p c / sigma\_T$$

Direct inputs (R)	Mass and plasma-composition assumptions
Inferred output (E)	Radiative-force balance scale
Validity boundary	Steady spherical ionized hydrogen approximation
Physical channel	Luminosity + spectrum
UM mapping	An environment-gated accretion scale with explicit assumptions.
Grade	E
Source	Open source

OM-015 — Chirp mass

$$M\_c = (m1\ m2)^{(3/5)} / (m1+m2)^{(1/5)}$$

Direct inputs (R)	Gravitational-wave phase evolution
Inferred output (E)	Best-measured binary mass combination
Validity boundary	Waveform family, detector calibration and source priors
Physical channel	Gravitational waves
UM mapping	A familiar waveform class carried without atomic-composition information.
Grade	E
Source	Open source

OM-016 — Angular gravitational scale

$$\text{theta\_g} = G\,M / (D\,c\text{\^{}}2)$$

Direct inputs (R)	Interferometric structure plus distance calibration
Inferred output (E)	Mass-to-distance or horizon-scale consistency
Validity boundary	Emission model, scattering, GRMHD calibration and geometry
Physical channel	Very-long-baseline interferometry
UM mapping	Separates measured plasma ring from inferred spacetime scale.
Grade	E
Source	Open source

OM-017 — Transmission Kernel

$$\text{K\^{}T\_S} = \{\text{Delta X} : \text{M\_S(X+Delta X)=M\_S(X)}\}$$

Direct inputs (R)	Declared measurement map and equivalence test
Inferred output (E)	World-state distinctions absent from the record
Validity boundary	Algebraic kernel only for linearized maps
Physical channel	Any channel
UM mapping	Formalizes the actual physical blind region of a chosen channel.

Grade	U
Source	UM_031 local artifact

OM-018 — Familiarity Kernel

$$K^F_S = \{d \text{ in } D_S : C_S(d)=\text{unfamiliar}\}$$

Direct inputs (R)	Record and current validated decoder
Inferred output (E)	Recorded patterns lacking a language assignment
Validity boundary	Not generally linear; depends on conservative class library
Physical channel	Any channel
UM mapping	Formalizes the familiarity insight without calling recorded structure inaccessible.
Grade	U
Source	UM_031 local artifact

OM-019 — Joint-channel blind region

$$K^T_{joint} = \text{intersection}_i\ K^T_i$$

Direct inputs (R)	Multiple calibrated measurement maps
Inferred output (E)	Distinctions erased by every included channel
Validity boundary	Linearized statement; correlated systematics remain
Physical channel	Multi-messenger
UM mapping	Shows why adding genuinely independent channels can reveal a previously hidden class.
Grade	E/U
Source	UM_031 local artifact

OM-020 — Four-clock age tuple

$$tau = (tau\_signal, tau\_formation, tau\_material, tau\_processing)$$

Direct inputs (R)	Distance, chronometers, population and process records
Inferred output (E)	Non-collapsed age description
Validity boundary	Each component has its own model and uncertainty
Physical channel	Multi-channel
UM mapping	Prevents 'light age' from being mistaken for matter or formation age.
Grade	U
Source	UM_030 local artifact

OM-021 — Complexity profile

$$kappa = (Diversity, Nesting, Retention, Channels, ModelDepth, Uncertainty)$$

Direct inputs (R)	Declared rubric scores with evidence notes
Inferred output (E)	Scale-relative descriptive vector
Validity boundary	Scores are ordinal and purpose-specific
Physical channel	Ledger synthesis
UM mapping	Represents inferred complexity without claiming a universal scalar essence.
Grade	U
Source	UM_032 local artifact

OM-022 — Environment-gated scalar example

$$V\_eff(s) = V0 - mu^2 s^2/2 + lambda s^4/4 + rho s^2/(2M^2)$$

Direct inputs (R)	Hypothetical parameters plus local density
Inferred output (E)	Screened/activated scalar response

Validity boundary	Candidate UM branch only; must satisfy laboratory, Solar-System and cosmological bounds
Physical channel	Proposed new dynamics
UM mapping	A possible mechanism for nature dependence, not part of the established ledger until predicted and tested.
Grade	U-hypothesis
Source	UM_029 local artifact

PART VII — COMPLEXITY PROFILES

The profile is ordinal, scale-relative, and purpose-specific. A score of five means high within this ledger, not maximal in the universe. The Navigation Index averages diversity, nesting, retention, channel diversity, and model depth. Uncertainty is kept separate.

7.1 Category counts

Class	Profiles
A-type main-sequence star	1
Barred spiral galaxy	2
Binary-merger black-hole remnant	1
Dwarf irregular galaxy	1
Element	5
G-type main-sequence star	1
Galactic-centre supermassive black hole	1
Gas giant	2
Giant elliptical galaxy	1
Ice giant	2
M-type red dwarf	1
Millisecond pulsar / neutron star	1
Red supergiant	1
Rocky ocean planet	1
Rocky planet	3
Starburst galaxy	1
Stellar-mass black-hole binary	1
Supermassive black hole	1
White dwarf	1

7.2 Full numeric profile matrix

ID	Entity	Class	Diversity	Nesting	Retention	Channels	Model depth	Uncertainty	Nav. index
EL-001	Hydrogen	Element	2	3	5	4	3	1	3.4
EL-006	Carbon	Element	5	4	5	5	4	2	4.6
EL-026	Iron	Element	4	4	5	5	4	2	4.4
EL-092	Uranium	Element	4	4	4	5	4	3	4.2
EL-118	Oganesson	Element	3	4	1	2	5	5	3
PL-001	Mercury	Rocky planet	3	4	5	4	3	2	3.8
PL-002	Venus	Rocky planet	4	4	5	4	4	3	4.2
PL-003	Earth	Rocky ocean planet	5	5	5	5	5	2	5
PL-004	Mars	Rocky planet	4	4	5	5	4	2	4.4
PL-005	Jupiter	Gas giant	5	5	5	5	5	3	5
PL-006	Saturn	Gas giant	5	5	5	5	5	3	5
PL-007	Uranus	Ice giant	4	5	5	3	5	4	4.4
PL-008	Neptune	Ice giant	4	5	5	3	5	4	4.4
ST-001	Sun	G-type main-sequence star	5	5	5	5	5	1	5
ST-002	Proxima Centauri	M-type red dwarf	4	5	5	4	4	3	4.4
ST-003	Sirius A	A-type main-sequence star	4	5	4	4	4	3	4.2
ST-004	Betelgeuse	Red supergiant	5	5	3	5	5	4	4.6
ST-005	Sirius B	White dwarf	3	5	5	4	5	3	4.4
ST-006	PSR J0740+6620	Millisecond pulsar / neutron star	3	5	5	5	5	4	4.6
GA-001	Milky Way	Barred spiral galaxy	5	5	5	5	5	3	5
GA-002	Andromeda / M31	Barred spiral galaxy	5	5	5	5	5	3	5
GA-003	M87	Giant elliptical galaxy	5	5	5	5	5	4	5

ID	Entity	Class	Diversity	Nesting	Retention	Channels	Model depth	Uncertainty	Nav. index
GA-004	Small Magellanic Cloud	Dwarf irregular galaxy	5	5	5	4	5	4	4.8
GA-005	M82	Starburst galaxy	5	5	4	5	5	4	4.8
BH-001	Sagittarius A*	Galactic-centre supermassive black hole	3	5	5	5	5	4	4.6
BH-002	M87*	Supermassive black hole	3	5	5	5	5	4	4.6
BH-003	Cygnus X-1	Stellar-mass black-hole binary	4	5	5	5	5	3	4.8
BH-004	GW150914 remnant	Binary-merger black-hole remnant	2	4	5	4	5	4	4

7.3 Interpretive notes

ID	Entity	Interpretive note
EL-001	Hydrogen	Simple atomic boundary; rich spectral and astrophysical roles.
EL-006	Carbon	Large bonding language and many environment-dependent molecular classes.
EL-026	Iron	Many ionization states, magnetic/solid phases and nucleosynthetic uses.
EL-092	Uranium	Radioactive clocks add nuclear-history channels.
EL-118	Oganesson	Short retention and sparse records make familiarity heavily model-dependent.
PL-001	Mercury	Nested atmosphere/interior/orbit complexity; channel coverage varies strongly by mission history.
PL-002	Venus	Nested atmosphere/interior/orbit complexity; channel coverage varies strongly by mission history.
PL-003	Earth	Nested atmosphere/interior/orbit complexity; channel coverage varies strongly by mission history.
PL-004	Mars	Nested atmosphere/interior/orbit complexity; channel coverage varies strongly by mission history.
PL-005	Jupiter	Nested atmosphere/interior/orbit complexity; channel coverage varies strongly by mission history.
PL-006	Saturn	Nested atmosphere/interior/orbit complexity; channel coverage varies strongly by mission history.
PL-007	Uranus	Nested atmosphere/interior/orbit complexity; channel coverage varies strongly by mission history.

ID	Entity	Interpretive note
PL-008	Neptune	Nested atmosphere/interior/orbit complexity; channel coverage varies strongly by mission history.
ST-001	Sun	Fusion, transport, magnetic, orbital and evolutionary clocks are partly reconstructed from boundary light.
ST-002	Proxima Centauri	Fusion, transport, magnetic, orbital and evolutionary clocks are partly reconstructed from boundary light.
ST-003	Sirius A	Fusion, transport, magnetic, orbital and evolutionary clocks are partly reconstructed from boundary light.
ST-004	Betelgeuse	Fusion, transport, magnetic, orbital and evolutionary clocks are partly reconstructed from boundary light.
ST-005	Sirius B	Fusion, transport, magnetic, orbital and evolutionary clocks are partly reconstructed from boundary light.
ST-006	PSR J0740+6620	Fusion, transport, magnetic, orbital and evolutionary clocks are partly reconstructed from boundary light.
GA-001	Milky Way	Population-level object with integrated-light loss, projection loss and strong multi-model inference.
GA-002	Andromeda / M31	Population-level object with integrated-light loss, projection loss and strong multi-model inference.
GA-003	M87	Population-level object with integrated-light loss, projection loss and strong multi-model inference.
GA-004	Small Magellanic Cloud	Population-level object with integrated-light loss, projection loss and strong multi-model inference.
GA-005	M82	Population-level object with integrated-light loss, projection loss and strong multi-model inference.
BH-001	Sagittarius A*	Compact external descriptors coexist with a deep, channel-dependent inferential chain.
BH-002	M87*	Compact external descriptors coexist with a deep, channel-dependent inferential chain.
BH-003	Cygnus X-1	Compact external descriptors coexist with a deep, channel-dependent inferential chain.
BH-004	GW150914 remnant	Compact external descriptors coexist with a deep, channel-dependent inferential chain.

PART VIII — COSMOLOGY OBSERVATIONAL LEDGER

COSMOLOGY RULE

Records first, established inference second, UM mapping third. Coverage is not confirmation.

The cosmology ledger separates direct probe records from model-propagated quantities. It preserves calibration, nuisance models, epoch boundaries, ensemble status, generic baselines, open tensions, and whether any independent UM prediction or held-out win exists.

8.1 Probe-family distribution

Probe family	Rows
21 cm reionization	1
Atom-interferometry fifth force	1
BBN baryon abundance	1
BBN radiation content	1
CMB acoustic scale	1
CMB background inference	1
CMB baryon loading	1
CMB large-scale polarization	1
CMB spectral distortion	1
CMB-inferred late growth	1
Cepheid-SN Ia distance ladder	1
Equivalence principle	1
Evolving dark-energy comparison	1
Executed standard-history interpreter	1
Galaxy and quasar BAO	1
Galaxy clustering + weak lensing	1
Gravitational-wave standard sirens	1
Hydrodynamic structure catalogue	1
Independent CMB spectra	1
Joint dark-energy fit	1
Lyman-alpha forest BAO	1
Neutrino mass suppression	1

Probe family	Rows
Optical cluster abundance	1
Primordial non-Gaussianity	1
Primordial scalar spectrum	1
Primordial tensor modes	1
Relativistic energy density	1
Type Ia supernova expansion	1
X-ray cluster abundance	1

8.2 Complete 29-row evidence matrix

COS-001 — CMB background inference

Epoch / redshift	z about 1100 -> today
Physical record (R)	Planck TT, TE, EE, low-l polarization and lensing spectra
Established inference (E)	Flat LambdaCDM inference of the present expansion rate
Quantity / value	H0: 67.4 km/s/Mpc; 68%; σ/limit 0.5
Dominant channel	Microwave anisotropy
Actual Boundary	Last-scattering opacity surface plus model evolution
Transmission limit	Foreground cleaning, beam/calibration and model degeneracy
Familiarity gap	No unexplained record class required in base fit
Ensemble status	Full-sky harmonic ensemble
UM mapping (U)	Signal record is separated from the model-propagated late-time clock
Generic baseline	Standard CMB likelihood
Core coverage	1
Independent UM prediction	0
Withheld win	0
Scalar relevance	Background + perturbations
Scalar ready / impact	0; Potential constraint
Internal verdict	Compatible with core interpretation; no UM-specific gain
Audit note	Model-dependent late-time inference, not a direct local H0 measurement.

Source	Open source

COS-002 — Independent CMB spectra

Epoch / redshift	z about 1100 -> today
Physical record (R)	ACT DR6 arcminute-scale TT, TE and EE with Planck large scales and lensing
Established inference (E)	Independent LambdaCDM expansion and initial-condition fit
Quantity / value	H0: 68.43 km/s/Mpc; 68% with DESI DR2; σ/limit 0.27
Dominant channel	Microwave anisotropy
Actual Boundary	Last scattering and ACT angular/foreground window
Transmission limit	Foregrounds, calibration and parameter degeneracies
Familiarity gap	No excess-lensing class required
Ensemble status	10,000 deg2 power-spectrum ensemble
UM mapping (U)	Transfers the same record/inference split to a different CMB instrument
Generic baseline	Standard CMB likelihood
Core coverage	1
Independent UM prediction	0
Withheld win	0
Scalar relevance	Background + perturbations
Scalar ready / impact	0; Potential constraint
Internal verdict	Cross-instrument consistency pass; not UM confirmation
Audit note	ACT reports no departure from flatness and no excess lensing.
Source	Open source

COS-003 — CMB baryon loading

Epoch / redshift	z about 1100
Physical record (R)	Relative acoustic peak amplitudes and polarization
Established inference (E)	Physical baryon density
Quantity / value	Omega_b h^2: 0.0224 68%; σ/limit 0.0001

Dominant channel	Microwave anisotropy
Actual Boundary	Photon-baryon acoustic plasma before decoupling
Transmission limit	Model dependence in recombination and foreground treatment
Familiarity gap	Established acoustic language is mature
Ensemble status	Multipole ensemble
UM mapping (U)	A retained matter coordinate inferred from propagation records
Generic baseline	LambdaCDM Boltzmann fit
Core coverage	1
Independent UM prediction	0
Withheld win	0
Scalar relevance	Early expansion
Scalar ready / impact	0; Potential constraint
Internal verdict	Mapped cleanly; numerical value not derived by UM
Audit note	Independent BBN comparison is recorded separately.
Source	Open source

COS-004 — CMB acoustic scale

Epoch / redshift	z about 1100
Physical record (R)	Angular positions of acoustic peaks
Established inference (E)	Sound-horizon angle at last scattering
Quantity / value	100 theta_*: 1.0411 68%; σ/limit 0.0003
Dominant channel	Microwave angular spectrum
Actual Boundary	Recombination visibility surface
Transmission limit	Absolute scale requires model and distance calibration
Familiarity gap	No unfamiliar peak family in the standard fit
Ensemble status	Full-sky angular modes
UM mapping (U)	A sharp transmission Boundary with a precise ensemble ruler
Generic baseline	Standard recombination + FLRW

Core coverage	1
Independent UM prediction	0
Withheld win	0
Scalar relevance	Background + recombination
Scalar ready / impact	0; Potential constraint
Internal verdict	Strong core example; no distinct UM prediction
Audit note	The measured angular scale is not itself a UM structural ratio.
Source	Open source

COS-005 — Primordial scalar spectrum

Epoch / redshift	Primordial -> z about 1100
Physical record (R)	CMB multipole dependence
Established inference (E)	Scalar spectral tilt
Quantity / value	n_s: 0.965 68%; σ/limit 0.004
Dominant channel	Microwave anisotropy
Actual Boundary	Primordial generation plus transfer functions
Transmission limit	Inflationary mechanism is not observed directly
Familiarity gap	Power-law class fits; detailed origin remains model-dependent
Ensemble status	Harmonic correlation ensemble
UM mapping (U)	Separates preserved spectrum from proposed generation history
Generic baseline	Power-law primordial LambdaCDM
Core coverage	1
Independent UM prediction	0
Withheld win	0
Scalar relevance	Initial conditions
Scalar ready / impact	0; Potential constraint
Internal verdict	Legacy r-power tilt claim remains quarantined
Audit note	UM has no registered primordial spectrum generator.

Source	Open source

COS-006 — CMB large-scale polarization

Epoch / redshift	Reionization
Physical record (R)	Low-l E-mode polarization
Established inference (E)	Integrated reionization optical depth
Quantity / value	tau: 0.054 68%; σ/limit 0.007
Dominant channel	Microwave polarization
Actual Boundary	Re-scattering during reionization
Transmission limit	Ionization-history degeneracy
Familiarity gap	Integrated tau does not uniquely recover reionization history
Ensemble status	Low-l sky modes
UM mapping (U)	Clear distinction between one integrated record and many histories
Generic baseline	Parameterized reionization
Core coverage	1
Independent UM prediction	0
Withheld win	0
Scalar relevance	Ionization history
Scalar ready / impact	0; Potential constraint
Internal verdict	Mapped; no UM-specific ionization prediction
Audit note	HERA provides a different reionization channel.
Source	Open source

COS-007 — Relativistic energy density

Epoch / redshift	BBN -> recombination
Physical record (R)	CMB damping tail plus BAO
Established inference (E)	Effective relativistic species
Quantity / value	N_eff: 2.99 68% with BAO; σ/limit 0.17

Dominant channel	Microwave + BAO
Actual Boundary	Thermal decoupling and radiation content
Transmission limit	Carrier identity is not fixed by N_eff alone
Familiarity gap	Extra-radiation class is familiar but undetected
Ensemble status	Joint cosmological likelihood
UM mapping (U)	A collective radiation coordinate, not a particle identification
Generic baseline	LambdaCDM + N_eff
Core coverage	1
Independent UM prediction	0
Withheld win	0
Scalar relevance	Early expansion
Scalar ready / impact	0; Potential constraint
Internal verdict	Legacy structural N_eff mapping remains unsupported
Audit note	BBN supplies an independent thermal-history comparison.
Source	Open source

COS-008 — Neutrino mass suppression

Epoch / redshift	Relic neutrino era -> today
Physical record (R)	CMB, lensing and DESI DR2 BAO
Established inference (E)	Upper limit on summed neutrino mass
Quantity / value	sum m_nu: 0.064 eV; 95% upper limit in LambdaCDM
Dominant channel	CMB + lensing + BAO
Actual Boundary	Free-streaming scale
Transmission limit	Mass hierarchy and model extensions remain degenerate
Familiarity gap	Standard neutrino response class
Ensemble status	Joint likelihood
UM mapping (U)	Scale-dependent retention/growth effect with explicit model boundary
Generic baseline	LambdaCDM + massive neutrinos

Core coverage	1
Independent UM prediction	0
Withheld win	0
Scalar relevance	Growth + background
Scalar ready / impact	0; Potential constraint
Internal verdict	Tight competing-growth constraint; no scalar solution tested
Audit note	Limit relaxes to 0.16 eV in w0waCDM.
Source	[Open source]

COS-009 — Primordial non-Gaussianity

Epoch / redshift	Primordial
Physical record (R)	Planck temperature and E-mode bispectra
Established inference (E)	Local bispectrum amplitude
Quantity / value	f_NL local: -0.9 68%; σ/limit 5.1
Dominant channel	Microwave three-point statistics
Actual Boundary	Primordial statistics propagated through last scattering
Transmission limit	Template choice and foreground residuals
Familiarity gap	No primordial NG detection
Ensemble status	Bispectrum estimator ensemble
UM mapping (U)	Fluctuation correlations extend beyond mean-field description
Generic baseline	Gaussian primordial LambdaCDM
Core coverage	1
Independent UM prediction	0
Withheld win	0
Scalar relevance	Initial conditions
Scalar ready / impact	0; Potential constraint
Internal verdict	Consistent mapping; no UM non-Gaussian prediction
Audit note	Null result constrains generation mechanisms.

Source	Open source

COS-010 — Primordial tensor modes

Epoch / redshift	Primordial
Physical record (R)	BICEP/Keck B-mode polarization with Planck and WMAP
Established inference (E)	Tensor-to-scalar upper limit
Quantity / value	r_0.05: 0.036 95% upper limit
Dominant channel	Microwave polarization
Actual Boundary	Foreground-separated B-mode window
Transmission limit	Dust and lensing separation limits primordial recovery
Familiarity gap	Primordial tensor class remains undetected
Ensemble status	Multi-frequency B-mode spectra
UM mapping (U)	A non-detection remains a bounded record, not absence of all tensors
Generic baseline	Inflationary tensor templates
Core coverage	1
Independent UM prediction	0
Withheld win	0
Scalar relevance	Initial conditions
Scalar ready / impact	0; Potential constraint
Internal verdict	No UM tensor prediction to compare
Audit note	Upper limit only.
Source	Open source

COS-011 — BBN baryon abundance

Epoch / redshift	Minutes after hot Big Bang
Physical record (R)	Primordial light-element abundance compilation and nuclear rates
Established inference (E)	Baryon density from nucleosynthesis
Quantity / value	Omega_b h^2: 0.02218 68% conservative; σ/limit 0.00055

Dominant channel	Spectroscopy + nuclear laboratory data
Actual Boundary	Nuclear reaction freeze-out
Transmission limit	Nuclear-rate and abundance systematics
Familiarity gap	Lithium remains anomalous; D and He broadly consistent
Ensemble status	Abundance + reaction-network ensemble
UM mapping (U)	Freeze-out is a rate-crossing retention Boundary
Generic baseline	Standard BBN
Core coverage	1
Independent UM prediction	0
Withheld win	0
Scalar relevance	Early expansion + trace
Scalar ready / impact	0; Potential constraint
Internal verdict	Independent agreement with CMB baryon density
Audit note	The agreement supports standard physics, not a UM numerical derivation.
Source	Open source

COS-012 — BBN radiation content

Epoch / redshift	Minutes after hot Big Bang
Physical record (R)	Light-element abundances with marginalized nuclear rates
Established inference (E)	Additional relativistic energy density
Quantity / value	Delta N_eff: -0.1 68%; σ/limit 0.21
Dominant channel	Abundance spectroscopy + nuclear data
Actual Boundary	Weak and nuclear freeze-out
Transmission limit	Reaction-rate and abundance systematics
Familiarity gap	No extra-radiation detection
Ensemble status	Reaction-network likelihood
UM mapping (U)	Independent channel for the same radiation coordinate
Generic baseline	BBN + N_eff

Core coverage	1
Independent UM prediction	0
Withheld win	0
Scalar relevance	Early expansion
Scalar ready / impact	0; Potential constraint
Internal verdict	Cross-channel consistency; scalar branch uncomputed
Audit note	Consistent with the standard radiation sector.
Source	Open source

COS-013 — Galaxy and quasar BAO

Epoch / redshift	0.1 < z < 4.2
Physical record (R)	DESI DR2 clustering of more than 14 million galaxies and quasars
Established inference (E)	Distance-redshift relation and BAO scale
Quantity / value	BAO-CMB tension: 2.3 sigma; flat LambdaCDM comparison
Dominant channel	Optical spectroscopy
Actual Boundary	Survey selection and sound-horizon ruler
Transmission limit	Bias, reconstruction and fiducial-model dependence
Familiarity gap	BAO feature is established
Ensemble status	Large-scale two-point ensemble
UM mapping (U)	A stable collective ruler emerging from many galaxy records
Generic baseline	Standard BAO likelihood
Core coverage	1
Independent UM prediction	0
Withheld win	0
Scalar relevance	Background expansion
Scalar ready / impact	0; Potential constraint
Internal verdict	Core mapping transfers; baseline explains same feature
Audit note	Flat LambdaCDM remains a good fit, with mild BAO-CMB parameter tension.

Source	Open source

COS-014 — Lyman-alpha forest BAO

Epoch / redshift	z about 2-4
Physical record (R)	DESI DR2 quasar absorption-forest correlations
Established inference (E)	High-redshift radial and transverse BAO distances
Quantity / value	BAO detection: Published likelihood
Dominant channel	Quasar spectroscopy
Actual Boundary	Intergalactic absorption and survey window
Transmission limit	Continuum fitting, metal contamination and bias model
Familiarity gap	Lyman-alpha correlation language is calibrated
Ensemble status	Forest auto/cross-correlation ensemble
UM mapping (U)	Periodic-table hydrogen identity becomes a cosmic field tracer
Generic baseline	Standard Ly-alpha BAO
Core coverage	1
Independent UM prediction	0
Withheld win	0
Scalar relevance	Background + growth
Scalar ready / impact	0; Potential constraint
Internal verdict	Strong periodic-to-cosmological bridge; not unique to UM
Audit note	Uses multiple forest/quasar correlation functions.
Source	Open source

COS-015 — Type Ia supernova expansion

Epoch / redshift	$0.025 < z < 1.13$
Physical record (R)	DES five-year standardized multi-band supernova light curves
Established inference (E)	Constant-w dark-energy fit from supernovae alone
Quantity / value	w: -0.8 approx symmetric 68%; σ/limit 0.15

Dominant channel	Optical time-series photometry + host redshifts
Actual Boundary	Selection, standardization and calibration chain
Transmission limit	Absolute luminosity and population evolution
Familiarity gap	SN Ia class validated probabilistically
Ensemble status	1,829-SN Hubble diagram
UM mapping (U)	Signal age and expansion inference are kept distinct
Generic baseline	Flat wCDM supernova likelihood
Core coverage	1
Independent UM prediction	0
Withheld win	0
Scalar relevance	Background expansion
Scalar ready / impact	0; Potential constraint
Internal verdict	Acceleration strongly detected; no UM-specific w prediction
Audit note	Supernova data alone require acceleration above 5 sigma in flat LambdaCDM.
Source	Open source

COS-016 — Joint dark-energy fit

Epoch / redshift	Low to intermediate redshift
Physical record (R)	DES supernovae + Planck + SDSS BAO + DES 3x2pt
Established inference (E)	Constant dark-energy equation of state
Quantity / value	w: -0.941 68%; σ/limit 0.026
Dominant channel	Multi-probe
Actual Boundary	Combined selection and model likelihood
Transmission limit	Cross-survey systematics and model dependence
Familiarity gap	Cosmological-constant class remains within about 2 sigma
Ensemble status	Joint likelihood
UM mapping (U)	Independent channels shrink different Transmission Kernels
Generic baseline	wCDM joint fit

Core coverage	1
Independent UM prediction	0
Withheld win	0
Scalar relevance	Background + growth
Scalar ready / impact	0; Potential constraint
Internal verdict	Compatible with Lambda; does not select UM scalar
Audit note	Different SN samples affect evolving-dark-energy significance.
Source	Open source

COS-017 — Cepheid-SN Ia distance ladder

Epoch / redshift	Local universe
Physical record (R)	HST Cepheids, geometric anchors and calibrated SNe Ia
Established inference (E)	Local expansion rate
Quantity / value	H0: 73.04 km/s/Mpc; 68%; σ/limit 1.04
Dominant channel	Optical/NIR distance ladder
Actual Boundary	Calibration ladder and local-flow corrections
Transmission limit	Zero points, population effects and peculiar velocities
Familiarity gap	Distance-ladder classes are mature but systematics debated
Ensemble status	Calibrator + Hubble-flow SN ensemble
UM mapping (U)	A late/local estimator is not identical to CMB model inference
Generic baseline	Distance-ladder fit
Core coverage	1
Independent UM prediction	0
Withheld win	0
Scalar relevance	Local/background response
Scalar ready / impact	0; Potential constraint
Internal verdict	H0 tension recorded, not resolved by the core
Audit note	Compared with Planck in the Tension Register.

Source	Open source

COS-018 — Gravitational-wave standard sirens

Epoch / redshift	Local to moderate redshift
Physical record (R)	GW luminosity distances plus host-galaxy or counterpart redshifts
Established inference (E)	Independent H0 posterior
Quantity / value	H0: 68 km/s/Mpc; approx symmetric 68%; σ/limit 4.1
Dominant channel	Gravitational waves + galaxy/EM records
Actual Boundary	Localization and host-association volume
Transmission limit	Inclination-distance and incomplete-catalog degeneracy
Familiarity gap	Siren class established; posterior remains broad
Ensemble status	15 dark plus bright-siren combination
UM mapping (U)	Independent channel reduces shared optical Transmission limits
Generic baseline	Standard-siren cosmology
Core coverage	1
Independent UM prediction	0
Withheld win	0
Scalar relevance	GW propagation + background
Scalar ready / impact	0; Potential constraint
Internal verdict	Consistent with both early and local values at current precision
Audit note	Reported 68.0 +4.4/-3.8 km/s/Mpc.
Source	Open source

COS-019 — CMB-inferred late growth

Epoch / redshift	Recombination -> today
Physical record (R)	Planck primary CMB spectra and lensing
Established inference (E)	Late clustering combination
Quantity / value	S8: 0.832 approx 68%; σ/limit 0.013

Dominant channel	Microwave anisotropy + lensing
Actual Boundary	Growth model from last scattering to today
Transmission limit	Neutrino, baryon and gravity-model degeneracies
Familiarity gap	Standard growth class
Ensemble status	CMB likelihood
UM mapping (U)	Early record propagated through an assumed dynamical model
Generic baseline	LambdaCDM growth
Core coverage	1
Independent UM prediction	0
Withheld win	0
Scalar relevance	Growth + lensing
Scalar ready / impact	0; Potential constraint
Internal verdict	Reference anchor for late-growth comparisons
Audit note	S8 value derived from Planck Omega_m and sigma8 summary.
Source	Open source

COS-020 — Galaxy clustering + weak lensing

Epoch / redshift	z below about 1.5
Physical record (R)	DES Y3 positions, shapes and galaxy-galaxy lensing
Established inference (E)	Late-time clustering amplitude
Quantity / value	S8: 0.776 68%; σ/limit 0.017
Dominant channel	Optical imaging + redshifts
Actual Boundary	Survey mask, source bins and lensing kernels
Transmission limit	Photo-z, shear calibration, intrinsic alignments and baryons
Familiarity gap	Weak-lensing language is calibrated with nuisance models
Ensemble status	Three two-point functions
UM mapping (U)	A late-time field inferred from many distorted galaxy images
Generic baseline	LambdaCDM 3x2pt

Core coverage	1
Independent UM prediction	0
Withheld win	0
Scalar relevance	Growth + lensing
Scalar ready / impact	0; Potential constraint
Internal verdict	Mild Planck comparison tension; not a UM success
Audit note	Comparison significance is computed in the Tension Register.
Source	Open source

COS-021 — Optical cluster abundance

Epoch / redshift	z below about 1
Physical record (R)	About 16,000 DES Y3 redMaPPer clusters with lensing and clustering
Established inference (E)	Cluster-calibrated growth amplitude
Quantity / value	S8: 0.811 approx symmetric 68%; σ/limit 0.021
Dominant channel	Optical richness + weak lensing
Actual Boundary	Cluster selection and mass-observable relation
Transmission limit	Projection, mass calibration and baryonic selection
Familiarity gap	Cluster population class is model-calibrated
Ensemble status	Cluster-count likelihood
UM mapping (U)	Population-level field; one cluster cannot set cosmic growth
Generic baseline	LambdaCDM cluster abundance
Core coverage	1
Independent UM prediction	0
Withheld win	0
Scalar relevance	Growth + screening
Scalar ready / impact	0; Potential constraint
Internal verdict	Consistent with CMB and shows late-probe dependence
Audit note	Joint clusters + 3x2pt result.

Source	Open source

COS-022 — X-ray cluster abundance

Epoch / redshift	z below about 1.3
Physical record (R)	5,259 eROSITA clusters with weak-lensing mass calibration
Established inference (E)	Cluster-calibrated growth amplitude
Quantity / value	S8: 0.86 68%; σ/limit 0.01
Dominant channel	X-ray counts + optical weak lensing
Actual Boundary	X-ray selection and count-rate/mass relation
Transmission limit	Hydrostatic, selection and mass-calibration systematics
Familiarity gap	X-ray cluster class calibrated statistically
Ensemble status	All-sky cluster-count likelihood
UM mapping (U)	Independent physical channel for the same growth coordinate
Generic baseline	LambdaCDM cluster abundance
Core coverage	1
Independent UM prediction	0
Withheld win	0
Scalar relevance	Growth + screening
Scalar ready / impact	0; Potential constraint
Internal verdict	Late probes do not form one tension-free number
Audit note	Reports Omega_m=0.29 +0.01/-0.02 and sigma8=0.88 +/-0.02.
Source	Open source

COS-023 — Evolving dark-energy comparison

Epoch / redshift	Late expansion
Physical record (R)	DESI DR2 BAO + CMB + alternative supernova compilations
Established inference (E)	Preference for w0wa over LambdaCDM
Quantity / value	Preference range: 3.5 2.8-4.2 sigma depending on SN sample

Dominant channel	Multi-probe
Actual Boundary	Model comparison and combined survey systematics
Transmission limit	Parameterization and SN-sample dependence
Familiarity gap	Evolving-w class is familiar but not established
Ensemble status	Joint likelihood/model comparison
UM mapping (U)	A live tension is recorded without promoting it to mechanism
Generic baseline	LambdaCDM vs w0waCDM
Core coverage	1
Independent UM prediction	0
Withheld win	0
Scalar relevance	Background + perturbations
Scalar ready / impact	0; Potential constraint
Internal verdict	Interesting external opening; no UM scalar likelihood exists
Audit note	DESI alone does not establish a UM mechanism.
Source	Open source

COS-024 — 21 cm reionization

Epoch / redshift	z=7.9 and 10.4
Physical record (R)	HERA interferometric delay-spectrum power
Established inference (E)	Upper limits on reionization-era 21 cm fluctuations
Quantity / value	Delta21^2: 946.18 mK^2; 95% upper at z=7.9
Dominant channel	Low-frequency radio interferometry
Actual Boundary	Foreground wedge, instrument response and EoR window
Transmission limit	Foregrounds and residual systematics erase many modes
Familiarity gap	Cosmological 21 cm signal remains undetected
Ensemble status	Visibility/power-spectrum ensemble
UM mapping (U)	A clean example of upper-limit and Transmission-Kernel discipline
Generic baseline	Standard EoR power-spectrum pipeline

Core coverage	1
Independent UM prediction	0
Withheld win	0
Scalar relevance	Reionization + thermal history
Scalar ready / impact	0; Potential constraint
Internal verdict	Coverage pass; no detection and no UM prediction
Audit note	(30.76 mK)^2 at k=0.192 h/Mpc.
Source	Open source

COS-025 — CMB spectral distortion

Epoch / redshift	Early-universe energy release
Physical record (R)	COBE/FIRAS absolute spectrum reanalysis
Established inference (E)	Upper limit on mean mu-type distortion
Quantity / value	abs(mu): 4.7e-05 95% upper limit
Dominant channel	Absolute microwave spectroscopy
Actual Boundary	Spectrometer calibration and foreground separation
Transmission limit	Limited FIRAS sensitivity and foreground modeling
Familiarity gap	Standard distortion templates; no robust detection
Ensemble status	Frequency-spectrum likelihood
UM mapping (U)	A null channel still constrains hidden energy processing
Generic baseline	Blackbody + distortion templates
Core coverage	1
Independent UM prediction	0
Withheld win	0
Scalar relevance	Early energy injection
Scalar ready / impact	0; Potential constraint
Internal verdict	No UM-specific energy-release calculation
Audit note	Useful null constraint on early new physics.

Source	Open source

COS-026 — Equivalence principle

Epoch / redshift	Present laboratory orbit
Physical record (R)	Differential acceleration of titanium and platinum test masses
Established inference (E)	Eotvos ratio consistent with zero
Quantity / value	eta(Ti,Pt): -1.5e-15 1 sigma; stat and syst combined approximately; σ/limit 2.75e-15
Dominant channel	Drag-free satellite accelerometry
Actual Boundary	Composition pair and orbital modulation
Transmission limit	Screening and common-mode systematics
Familiarity gap	WEP-violation template is familiar and null
Ensemble status	Five months of free-fall data
UM mapping (U)	Local composition record directly tests any non-universal response
Generic baseline	General relativity / WEP
Core coverage	1
Independent UM prediction	0
Withheld win	0
Scalar relevance	Visible-sector universality
Scalar ready / impact	0; Potential constraint
Internal verdict	Strong guardrail; not support for the scalar
Audit note	Published eta=[-1.5 +/-2.3(stat) +/-1.5(syst)]e-15.
Source	Open source

COS-027 — Atom-interferometry fifth force

Epoch / redshift	Present laboratory
Physical record (R)	Atomic acceleration measurements in screened laboratory geometry
Established inference (E)	Excluded symmetron parameter regions
Quantity / value	Parameter exclusion: Model-dependent

Dominant channel	Atom interferometry
Actual Boundary	Vacuum chamber, source mass and atomic trajectory
Transmission limit	Geometry and nonlinear screening solution
Familiarity gap	Symmetron acceleration template is established
Ensemble status	Repeated atom-interference measurements
UM mapping (U)	Direct nature-dependent gate test at laboratory scale
Generic baseline	Symmetron/chameleon fifth-force models
Core coverage	1
Independent UM prediction	0
Withheld win	0
Scalar relevance	Activation + screening
Scalar ready / impact	0; Potential constraint
Internal verdict	UM fixed ratio has not been propagated through this experiment
Audit note	A numerical UM point or region is still required.
Source	Open source

COS-028 — Executed standard-history interpreter

Epoch / redshift	Radiation era -> today
Physical record (R)	1,200-row supplied CAMB/Planck background history
Established inference (E)	Equality, recombination and acceleration milestones
Quantity / value	z_eq: 3468 fractional grid error stored in sigma column; σ/limit 0.0007076
Dominant channel	Simulation state history
Actual Boundary	Declared background-component equalities
Transmission limit	No perturbation or object catalogue in this record
Familiarity gap	Standard milestones are fully familiar
Ensemble status	Dense time-grid history
UM mapping (U)	Downstream UM labels with unchanged physical evolution
Generic baseline	Generic equality and entropy markers

Core coverage	1
Independent UM prediction	0
Withheld win	0
Scalar relevance	None
Scalar ready / impact	0; Not applicable
Internal verdict	Implementation pass; generic baselines peak identically
Audit note	This is the strongest executed internal null comparison.
Source	[Open source]()

COS-029 — Hydrodynamic structure catalogue

Epoch / redshift	Galaxy formation history
Physical record (R)	TNG subhalo physical fields through a frozen adapter
Established inference (E)	Potential, activity, metallicity and environment coordinates
Quantity / value	Real-catalogue run: Pending
Dominant channel	Simulation catalogue
Actual Boundary	Subhalo finder, resolution and aperture definitions
Transmission limit	Merger history absent without trees; catalogue not supplied
Familiarity gap	TNG fields are familiar; UM transfer advantage untested
Ensemble status	Population catalogue
UM mapping (U)	Independent object mapping without percentile-forced fractions
Generic baseline	Density, clustering and generic two-axis mappings
Core coverage	1
Independent UM prediction	0
Withheld win	0
Scalar relevance	None
Scalar ready / impact	0; Not applicable
Internal verdict	Interface passes; no real-catalogue evidence yet
Audit note	This is the next direct core-compiler comparison.

Source	Open source

8.3 Tension register

ID	Issue	Input records	Strength / metric	Status	Core UM reading	Optional scalar status	Decisive next test	Sources
-001	H0: Planck vs SH0ES	COS-001 and COS-017	4.887561225309663	Open	Different channels and inference chains disagree; the core records the estimator Boundary but supplies no shift mechanism.	Uncomputed; any scalar solution must fit CMB, BAO and local screening together.	Independent ladders, sirens and a frozen joint likelihood.	COSRC-001; COSRC-00
-002	H0: ACT+DESI vs SH0ES	COS-002 and COS-017	4.290460981672407	Open	The tension persists with an independent high-resolution CMB chain.	Uncomputed.	Same joint-likelihood test with ACT nuisance models.	COSRC-002; COSRC-00
-003	S8: Planck vs DES Y3	COS-019 and COS-020	2.6167081556895844	Open	A mild early-to-late growth mismatch; not a uniform late-probe result.	Potentially relevant but no nonlinear prediction exists.	Frozen multi-probe growth and lensing likelihood.	COSRC-001; COSRC-00
-004	S8: DES Y3 vs eROSITA clusters	COS-020 and COS-022	4.258969063132506	Open	Late probes themselves differ, emphasizing calibration and selection Boundaries.	Screening could affect clusters, but no fitted profile exists.	Common mass calibration and selection-controlled cross-analysis.	COSRC-008; COSRC-01
-005	Evolving dark energy	COS-023	2.8-4.2 sigma, SN-sample dependent	Open	A legitimate external anomaly; the Universopedia prevents it being mistaken for a UM prediction.	Relevant, but the frozen scalar has no background/perturbation likelihood.	DESI full survey plus multiple supernova calibrations.	COSRC-003; COSRC-00
-006	Planck excess-lensing preference vs ACT	COS-001 and COS-002	>2 sigma in Planck internal spectra; absent in ACT DR6	Open	Cross-instrument Familiarity test suggests caution about promoting an internal anomaly.	Constrains growth modifications.	Joint map/likelihood reanalysis with common foreground treatment.	COSRC-001; COSRC-00
-007	Primordial lithium problem	COS-011	D and He broadly agree; Li-7 remains discrepant	Open	The periodic language identifies the isotope record but not its missing mechanism.	Early expansion is constrained; no UM abundance network exists.	Improved stellar depletion and nuclear-rate evidence.	COSRC-013
-008	Reionization 21 cm signal	COS-006 and COS-024	Upper limit only	Open	CMB tau is an integrated record; HERA exposes different modes and systematics.	No thermal/reionization prediction exists.	Deeper HERA/SKA detections with cross-correlation.	COSRC-001; COSRC-01
-009	UM scalar viability region	COS-003 to COS-027	No joint numerical region or likelihood	Unexecuted	Not part of the core interpreter.	This is the remaining optional-physics calculation, including activation history and domain walls.	Solve background, perturbations and nonlinear screening before fitting.	COSRC-016; COSRC-01

8.4 Comparison verdict

CURRENT COSMOLOGY VERDICT

Coverage pass / distinctiveness not yet shown / optional scalar not yet quantitatively tested.

- All 29 audited records preserve R, E, U, Boundary, Transmission, and source separation.
- No row contains a preregistered independent UM numerical prediction or a held-out win.
- Open external tensions remain recorded rather than converted into UM successes.

- The ledger demonstrates breadth and auditability, not physical confirmation.

PART IX — ILLUSTRISTNG EMPIRICAL PROGRAMME

9. Prospective core test

PRIMARY QUESTION
Does a fixed UM compiler predict withheld metallicity, gas, host, central/satellite, and environmental records better than information-matched and generic baselines, without retuning across resolution, volume, and epoch transfers?

The TNG test leaves the simulation dynamics unchanged. UM is a downstream compiler. The decisive object is not the entire simulation archive; it is a frozen adapter, a limited set of catalogue fields, protected targets, checksums, fixed partitions, and fair baseline comparisons.

9.1 Frozen decisions

ID	Decision	Frozen rule	Why necessary	Status	May change?
D01	Physical dynamics	Use TNG outputs unchanged; UM is downstream only.	Prevents descriptive success being reported as altered physics.	FROZEN	NO
D02	Independent mapping	Map every galaxy independently; never sort and cut to desired UM fractions.	Prevents the output distribution being forced by construction.	FROZEN	NO
D03	Adapter inputs	Only the two original adapter channels may enter: star-formation activity over a Hubble time and potential depth relative to the atomic-cooling velocity.	Preserves the documented frozen adapter concept.	AWAITING SOURCE HASH	NO
D04	Withheld targets	Gas metallicity, stellar metallicity, gas fraction, host-halo mass, central/satellite identity and spatial environment are unavailable to the adapter.	Creates genuine withheld records rather than relabelled inputs.	FROZEN	NO
D05	Eligibility	SubhaloFlag=1 and stellar mass >= 1e10 Msun; target-specific finite/positive requirements are applied only after the common sample is frozen.	Keeps a well-resolved common sample across the declared transfer runs.	PROPOSED FREEZE	NO
D06	Partition	All subhalos of one FoF host and one spatial block remain in the same train/validation/test partition.	Prevents same-halo and near-neighbour leakage.	FROZEN	NO
D07	Model fairness	UM-2 and Raw-2 use the same estimator family, cross-validation budget and complexity penalty.	The UM coordinates are a deterministic two-degree-of-freedom transform of the raw adapter inputs.	FROZEN	NO
D08	Transfer	Fit on TNG100-1 snapshot 99 only. Transfer catalogues receive no refit and no threshold retuning.	Makes resolution, volume and epoch transfer decisive.	FROZEN	NO
D09	Nulls	Run target shuffles, adapter-row shuffles, catalogue-order permutations and unit-rescaling tests.	Separates real association from implementation artifacts.	FROZEN	NO
D10	Publication rule	A null or baseline-equivalent result is recorded as the result.	Prevents the core language test from becoming confirmation-only.	FROZEN	NO

9.2 Model families

Model	Feature family	Inputs	DOF	Role	Decisive comparison
M0	Null	Training-set median / class prevalence	0	Sanity baseline	Every model must beat this on held-out records.
M1	Raw-2	Two raw adapter coordinates before the UM quadratic map	2	Information-matched baseline	M2 versus M1 tests representation/compression, not creation of new information.
M2	UM-2	Two independent UM weights; the third is omitted because W_L+W_B+W_M=1	2	Primary UM representation	Primary transfer comparison against M1 using identical estimator complexity.

Model	Feature family	Inputs	DOF	Role	Decisive comparison
M3	Generic physical	Stellar mass, raw activity, Vmax, central flag and spatial density	5+	Generic density/clustering baseline	Tests whether UM adds anything beyond ordinary physical descriptors.
M4	Generic + UM	M3 plus two independent UM weights	7+	Incremental-value test	M4-M3 must improve held-out transfer after complexity penalty.

9.3 Primary gates

Gate	Pass rule	Primary statistic	Failure meaning
G2 — Resolution stability	Unchanged M2 mapping and fitted predictor retain performance on TNG100-2 within the predeclared uncertainty interval.	Paired spatial-block bootstrap of transfer loss.	The compact UM representation is resolution-sensitive.
G3 — Transfer	M2 transfers without retuning to at least two of resolution, volume and epoch catalogues.	Mean standardized transfer score across primary continuous targets.	The mapping does not generalize beyond its training catalogue.
G4 — Withheld prediction	M2 beats M0 and predicts at least two primary withheld continuous records with bootstrap interval excluding no skill.	Out-of-sample R^2 and RMSE.	The coordinates do not carry useful withheld structure.
G5 — Baseline advantage	M2 beats information-matched M1, or M4 improves over M3 after complexity penalty, on the preregistered transfer composite.	Paired loss difference and penalized score.	UM may remain descriptive but is not empirically distinctive.
Shuffle null	Observed M2 score exceeds 99% of row-shuffled scores.	Permutation p-value <= 0.01.	The apparent signal is compatible with accidental association.

9.4 Field map and leakage control

Field_ID	TNG field / derived record	Table	Units / shape	Role	Frozen transformation	Used by UM adapter?	Used as withheld target?	Leakage control	Official source
F001	HubbleParam	Header	dimensionless	Metadata	Mass conversion h and cosmological time metadata	NO	NO	Metadata only	https://www.tng-project.org/data/docs/specifications/
F002	Time	Header	scale factor	Metadata	a=Time; redshift read from Header or run manifest	NO	NO	Metadata only	https://www.tng-project.org/data/docs/specifications/
F003	BoxSize	Header	ckpc/h	Metadata	Periodic distances and spatial blocks	NO	NO	Never used for target value	https://www.tng-project.org/data/docs/specifications/

Field_ID	TNG field / derived record	Table	Units / shape	Role	Frozen transformation	Used by UM adapter?	Used as withheld target?	Leakage control	Official source
F004	SubhaloFlag	Subhalo	integer	Quality cut	Keep only cosmological-origin objects with flag=1	NO	NO	Eligibility fixed before targets	https://www.tng-project.org/data/docs/specifications/
F005	SubhaloMassType[:,4]	Subhalo	1e10 Msun/h	Quality + generic baseline	M_star = value*1e10/h; common cut M_star>=1e10 Msun	NO	NO	Not permitted in M2; permitted in M3/M4	https://www.tng-project.org/data/docs/specifications/
F006	SubhaloMassType[:,0]	Subhalo	1e10 Msun/h	Withheld gas record	M_gas = value*1e10/h; gas fraction formed after common sample freeze	NO	YES	Never passed to adapter or Raw-2	https://www.tng-project.org/data/docs/specifications/
F007	SubhaloSFR	Subhalo	Msun/yr	Adapter input	Original frozen activity-over-Hubble-time map; exact source required	YES	NO	Target columns unopened during mapping	https://www.tng-project.org/data/docs/specifications/
F008	SubhaloVmax	Subhalo	km/s	Adapter input	Original frozen potential-depth / atomic-cooling map; exact source required	YES	NO	Target columns unopened during mapping	https://www.tng-project.org/data/docs/specifications/
F009	SubhaloGasMetallicity	Subhalo	mass fraction	Primary withheld target	log10(Z_gas) for finite positive records	NO	YES	Target-specific valid mask after common sample	https://www.tng-project.org/data/docs/specifications/
F010	SubhaloStarMetallicity	Subhalo	mass fraction	Primary withheld target	log10(Z_star) for finite positive records	NO	YES	Target-specific valid mask after common sample	https://www.tng-project.org/data/docs/specifications/
F011	SubhaloPos	Subhalo	N x 3 ckpc/h	Environment record	Periodic neighbour distances and spatial block ID	NO	YES	Environment is absent from adapter inputs	https://www.tng-project.org/data/docs/specifications/
F012	SubhaloGrNr	Subhalo	group index	Split + environment	Join to FoF host; all members kept in one partition	NO	YES	Prevents same-host leakage	https://www.tng-project.org/data/docs/specifications/
F013	GroupFirstSub	Group	subhalo index	Environment target	Central=1 when subhalo index equals host GroupFirstSub	NO	YES	Not passed to adapter or Raw-2	https://www.tng-project.org/data/docs/specifications/
F014	Group_M_Crit200	Group	1e10 Msun/h	Primary environment target	log10(M_200c*1e10/h) for positive records	NO	YES	Host property withheld from adapter	https://www.tng-project.org/data/docs/specifications/
F015	Group_R_Crit200	Group	ckpc/h	Environment target support	Host-centric distance divided by R_200c	NO	YES	Target only; no adapter access	https://www.tng-project.org/data/docs/specifications/
F016	GroupPos	Group	N x 3 ckpc/h	Environment target support	Periodic host-centric distance	NO	YES	Target only; no adapter access	https://www.tng-project.org/data/docs/specifications/

Field_ID	TNG field / derived record	Table	Units / shape	Role	Frozen transformation	Used by UM adapter?	Used as withheld target?	Leakage control	Official source
F017	GroupNsubs	Group	count	Secondary environment target	log1p(host richness)	NO	YES	Target only; no adapter access	https://www.tng-project.org/data/docs/specifications/
F018	Derived local density	Derived	number / volume	Generic clustering baseline + target	k=5 periodic neighbour density within common mass-limited sample	NO	YES	Never used by M2; explicit M3/M4 only	https://www.tng-project.org/data/docs/specifications/
F019	UM weights W_L,W_B,W_M	Derived	three weights summing to 1	UM feature family	Computed only by the hash-verified frozen adapter	OUTPUT	NO	Drop one weight in regression to avoid exact collinearity	Local frozen adapter source

9.5 Run ledger

n_ID	Simulation	Snapshot	Role	Frozen fit source	Adapter SHA-256	Data SHA-256	N raw	N eligible	Split / seed	Status	Output manifest / notes
R01	TNG100-1	99	Training + internal held-out	This run					FoF+spatial / 241	WAITING FOR DATA	Fit M0-M4; freeze preprocessing, estimator and nuisance choices.
R02	TNG100-2	99	Resolution transfer	TNG-R01					All eligible	WAITING FOR DATA	No refit; unchanged adapter and fitted models.
R03	TNG300-1	99	Volume transfer	TNG-R01					All eligible	WAITING FOR DATA	No refit; tests volume and environment distributio shift.
R04	TNG100-1	67	Epoch transfer	TNG-R01					All eligible	WAITING FOR DATA	No refit or epoch-specific threshold adjustment.
R05	TNG100-1	99	Target shuffle null	TNG-R01					100 permutations	WAITING FOR DATA	Shuffle target association within the fixed held-out partition.
R06	Synthetic fixture	N/A	Unit-rescaling invariance	None		generated			deterministic	READY FOR HARNESS	Re-run common mass and SFR rescalings; expecte numerical invariance.
R07	Synthetic fixture	N/A	Catalogue-order invariance	None	1443bb9f93161e9a18c3222a346a1cb98347d64db05e2f185bd498c81ea22bd9	048b0254671f079fc497495839585e4ce846f81ef62a5d72473d674c4fd4bb6b	512	438	deterministic	HARNESS PASS — DUMMY ONLY	Synthetic order-invariance error 0.0; closure error 2 16. Plumbing only—not the frozen UM adapter an a TNG result.
L01	Published TNG literature	Multiple	Retrospective summary comparison	None	N/A	Paper URLs + quoted anchors	9	26	No split / no fitting	COMPLETE — SUMMARY ONLY	Nine primary papers, 26 evidence statements. No catalogue objects, adapter fit, target leakage, or superiority claim.

9.6 Predeclared results grid

The results grid contains 120 predeclared cells: 4 primary catalogues × 6 targets × 5 model families. Blank metric cells mean the authenticated real-catalogue run has not occurred; they are not failures and must not be back-filled from retrospective literature.

Dimension	Declared values
Runs	TNG-R01 training/internal held-out; TNG-R02 resolution; TNG-R03 volume; TNG-R04 epoch.
Targets	log10 gas metallicity; log10 stellar metallicity; gas fraction; log10 host M200c; central/satellite; local density.
Models	M0, M1 Raw-2, M2 UM-2, M3 generic physical, M4 generic + UM.
Metrics	N, RMSE, MAE, R^2, Spearman, bootstrap interval, delta versus Raw-2.

9.7 Current execution boundary

- The exact earlier adapter formula and SHA-256 were not found in the supplied source set; the adapter contract was intentionally left empty rather than guessed.
- Authenticated catalogue access was pending at the time of this edition.
- Synthetic unit, closure, order-invariance, hashing, and leakage-guard tests passed; they validate plumbing only.
- The real object-level verdict remains open, not failed.

9.8 TNG literature phase

RETROSPECTIVE VERDICT

LANGUAGE SURVIVES; SUPERIORITY UNTESTED.

Nine primary papers supplied 26 published statements across metallicity, gas, environment, quenching, structure, clustering, and data access. The phase checked whether the frozen language could reconstruct the published relations without a glaring contradiction. It performed no fitting, no catalogue-level prediction, and no superiority test.

9.9 Literature interpretation rules

Rule	Frozen rule	Purpose	May change?	Status	Failure condition
LR01	Published statements are transcribed with numeric anchors and source URLs.	Traceability	NO	FROZEN	Missing or altered source statement
LR02	The phase is labelled retrospective in every outcome summary.	Prevents artificial confirmation	NO	FROZEN	Any prospective-prediction wording
LR03	TNG-vs-observation tensions receive neutral UM scores and remain visible.	Prevents cherry-picking	NO	FROZEN	A source tension counted as UM success
LR04	Compatibility scores assess language coverage only, not causal truth.	Separates interpretation from dynamics	NO	FROZEN	Score described as physical confirmation
LR05	Mass-only, environment-only, and generic astrophysical baselines remain explicit.	Tests whether UM adds organizational language	NO	FROZEN	Baseline omitted from a distinctive row

Rule	Frozen rule	Purpose	May change?	Status	Failure condition
LR06	The full object-level held-out test remains the decisive core test.	Preserves the empirical ladder	NO	FROZEN	Literature phase presented as replacement

9.10 Complete 26-statement literature ledger

LIT-001 — Metallicity (P02)

Simulation / sample	TNG100
Scope	0<z<2; 10^9<M*<10^10.5 Msun
Published relation	TNG broadly reproduces the slope and normalization evolution of the gas-phase mass-metallicity relation.
Numeric anchor	MZR reproduced across stated mass/redshift range
Target record	Gas-phase metallicity
Conditioning variables	Stellar mass; epoch
External status	Published TNG result
UM mapping	A stable ensemble relation links retained stellar assembly to a gas-phase record at a declared scale.
Generic baseline / missing structure	Stellar-mass-only description captures the mean relation.
Compatibility	Partial
Distinctive?	NO
Score	1
Claim impact	COMPATIBLE / NON-DISTINCTIVE
Caveat / interpretation	Compatible, but not distinctive: ordinary mass-metallicity language already captures the main trend.
Source	Open source

LIT-002 — Metallicity (P02)

Simulation / sample	TNG100
Scope	2<z<10
Published relation	The high-redshift MZR normalization gradually declines with increasing redshift.
Numeric anchor	d log(Z)/dz ≈ -0.064 for z>2
Target record	Gas-phase metallicity

Conditioning variables	Epoch; gas fraction
External status	Published TNG result
UM mapping	The record changes with processing age and the retained gas reservoir, rather than through a timeless object label.
Generic baseline / missing structure	A redshift-only trend records the direction but not the stated gas-fraction mechanism.
Compatibility	Partial
Distinctive?	NO
Score	1
Claim impact	COMPATIBLE / NON-DISTINCTIVE
Caveat / interpretation	Compatible and useful for the age/retention vocabulary, but not a unique UM result.
Source	Open source

LIT-003 — Metallicity (P02)

Simulation / sample	TNG100
Scope	z=0 global metal distribution
Published relation	Most gas-phase metals lie outside the star-forming ISM, in extended disks, CGM, or beyond the halo.
Numeric anchor	~85% of gas-phase metals outside the ISM
Target record	Metal mass by phase
Conditioning variables	Chosen physical boundary; gas phase
External status	Published TNG result
UM mapping	The actual Boundary is scale- and question-dependent: generated material can leave the local retained identity while remaining in larger nested systems.
Generic baseline / missing structure	A fixed 'galaxy equals ISM' boundary misses the dominant metal reservoir.
Compatibility	Strong
Distinctive?	YES
Score	2
Claim impact	SUPPORTIVE BUT RETROSPECTIVE
Caveat / interpretation	Directly instantiates nested Boundaries and exchange across a metastable object boundary.
Source	Open source

LIT-004 — Metallicity (P02)

Simulation / sample	TNG100
Scope	Fixed stellar mass
Published relation	MZR scatter correlates with gas mass or SFR and reflects competition between enrichment- and accretion-dominated phases.
Numeric anchor	Correlated residuals at fixed M*
Target record	Metallicity residual
Conditioning variables	Gas mass; SFR; accretion history
External status	Published TNG result
UM mapping	Update, retention, and exchanged material jointly determine the familiar metallicity record; one scalar mass coordinate is insufficient.
Generic baseline / missing structure	Mass-only fails by construction at fixed stellar mass.
Compatibility	Strong
Distinctive?	YES
Score	2
Claim impact	SUPPORTIVE BUT RETROSPECTIVE
Caveat / interpretation	A strong cross-channel relation: retention state and current update activity condition the record.
Source	Open source

LIT-005 — Metallicity (P06)

Simulation / sample	IllustrisTNG
Scope	z<0.5; 10^9<M*<10^10 Msun; future cluster members
Published relation	Galaxies that will later enter clusters are already more metal-rich than field galaxies before infall, driven by pre-enriched gas accretion.
Numeric anchor	~0.05 dex MZR enhancement before infall
Target record	Gas-phase metallicity
Conditioning variables	Future environment; enriched inflow; pre-infall history
External status	Published TNG result
UM mapping	The causal Boundary is not identical to the nominal virial shell; environmental records and processing history can act before geometric membership.

Generic baseline / missing structure	Current-host or inside/outside-only classification misses the pre-infall effect.
Compatibility	Strong
Distinctive?	YES
Score	2
Claim impact	SUPPORTIVE BUT RETROSPECTIVE
Caveat / interpretation	One of the clearest literature instantiations of the actual-Boundary and processing-age distinction.
Source	Open source

LIT-006 — Gas (P03)

Simulation / sample	IllustrisTNG
Scope	Primarily z≈0
Published relation	Simulated gas fractions and the HI mass-size relation generally agree well with observations.
Numeric anchor	Broad agreement; tight HI size-mass relation
Target record	HI/H2 mass and size
Conditioning variables	Stellar mass; morphology; aperture
External status	Published TNG result
UM mapping	Stable ensemble statistics support a repeatable gas-field description at a declared scale.
Generic baseline / missing structure	Standard gas scaling relations already describe this result.
Compatibility	Partial
Distinctive?	NO
Score	1
Claim impact	COMPATIBLE / NON-DISTINCTIVE
Caveat / interpretation	Useful consistency evidence, but not a discriminating UM test.
Source	Open source

LIT-007 — Gas (P03)

Simulation / sample	IllustrisTNG

Scope	0≤z≤2
Published relation	H2 mass correlates with SFR through an almost constant depletion time that evolves with redshift as observed.
Numeric anchor	Near-constant molecular depletion time to z=2
Target record	Molecular gas / SFR
Conditioning variables	Epoch; star formation activity
External status	Published TNG result
UM mapping	A retained molecular reservoir and its update rate form a stable relational clock rather than two unrelated labels.
Generic baseline / missing structure	Generic gas-regulator models already encode this relation.
Compatibility	Partial
Distinctive?	NO
Score	1
Claim impact	COMPATIBLE / NON-DISTINCTIVE
Caveat / interpretation	Compatible with retention/update semantics; not uniquely UM.
Source	Open source

LIT-008 — Gas (P03)

Simulation / sample	IllustrisTNG
Scope	z=0
Published relation	The simulation overestimates the total neutral-gas abundance compared with observations.
Numeric anchor	About a factor of 2 high
Target record	Cosmic neutral-gas abundance
Conditioning variables	Gas partition model; calibration
External status	TNG-vs-observation tension
UM mapping	This is retained as an external-model tension and is not converted into a UM success or failure.
Generic baseline / missing structure	No language framework removes a quantitative source-model discrepancy.
Compatibility	Neutral
Distinctive?	NO

Score	0
Claim impact	TNG TENSION — NOT UM FAILURE
Caveat / interpretation	Important limitation of the TNG evidence source.
Source	Open source

LIT-009 — Gas (P03)

Simulation / sample	IllustrisTNG
Scope	Low redshift satellites
Published relation	The simulation may contain too many satellites with little or no neutral gas.
Numeric anchor	Possible excess of gas-poor satellites
Target record	Satellite neutral gas
Conditioning variables	Satellite status; stripping model
External status	TNG-vs-observation tension
UM mapping	Recorded as a source-model caveat; it prevents treating every satellite gas trend as observational confirmation.
Generic baseline / missing structure	No interpretive baseline resolves the mismatch.
Compatibility	Neutral
Distinctive?	NO
Score	0
Claim impact	TNG TENSION — NOT UM FAILURE
Caveat / interpretation	Explicitly prevents confirmation-only reading of the literature.
Source	Open source

LIT-010 — Gas (P09)

Simulation / sample	TNG50
Scope	z=0; satellites within 300 kpc
Published relation	Gas content drops as satellites cross the host virial radius; most nearby satellites lack detectable gas unless unusually massive.
Numeric anchor	Most within 300 kpc gas-poor; massive exceptions

Target record	Satellite gas reservoir
Conditioning variables	Virial crossing; host distance; satellite mass
External status	Published TNG result
UM mapping	Crossing an environmental Boundary changes retention, but identity can persist; the effect is mass- and channel-dependent.
Generic baseline / missing structure	Distance-only or satellite-label-only descriptions miss the mass-dependent exception and crossing history.
Compatibility	Strong
Distinctive?	YES
Score	2
Claim impact	SUPPORTIVE BUT RETROSPECTIVE
Caveat / interpretation	Strong Boundary/retention mapping with an explicit exception class.
Source	Open source

LIT-011 — Gas (P09)

Simulation / sample	TNG50
Scope	Present-day quenched satellites
Published relation	Quenched satellites typically stopped forming stars long ago, although only a few percent visibly appear as jellyfish.
Numeric anchor	Time since quenching 6.9 (+2.5/-3.3) Gyr; few % jellyfish
Target record	Quenching history and morphology
Conditioning variables	Processing age; stripping history; visible morphology
External status	Published TNG result
UM mapping	Current appearance is an incomplete record of the causal history: processing age and Transmission limits separate invisible past stripping from visible morphology.
Generic baseline / missing structure	Morphology-only classification misses most of the inferred history.
Compatibility	Strong
Distinctive?	YES
Score	2
Claim impact	SUPPORTIVE BUT RETROSPECTIVE
Caveat / interpretation	Directly uses separate processing age and record-versus-history language.

Source	Open source

LIT-012 — Environment (P04)

Simulation / sample	TNG50/100/300
Scope	Satellites in 10^12–10^15.2 Msun hosts
Published relation	At fixed stellar mass, satellites have lower dynamical mass than centrals; at fixed dynamical mass, their stellar fraction is higher.
Numeric anchor	Stellar-to-total fraction higher by up to a factor of a few
Target record	Stellar and dynamical mass
Conditioning variables	Central/satellite identity; host environment
External status	Published TNG result
UM mapping	A metastable stellar identity can persist while outer retained mass crosses the environmental Boundary.
Generic baseline / missing structure	A single stellar-mass or halo-mass coordinate cannot preserve both conditional relations.
Compatibility	Strong
Distinctive?	YES
Score	2
Claim impact	SUPPORTIVE BUT RETROSPECTIVE
Caveat / interpretation	A strong whole/retained-component distinction.
Source	Open source

LIT-013 — Environment (P04)

Simulation / sample	TNG50/100/300
Scope	z<2
Published relation	The satellite-central SHMR difference grows in more massive hosts, at smaller radii, for earlier infall, and at higher local density.
Numeric anchor	Four reinforcing environmental conditions
Target record	SHMR offset
Conditioning variables	Host mass; radius; infall time; local density
External status	Published TNG result

UM mapping	Boundary exchange depends jointly on scale, environment, and retained history; no single present-day coordinate is sufficient.
Generic baseline / missing structure	Mass-only or clustering-only baselines each omit declared conditioning variables.
Compatibility	Strong
Distinctive?	YES
Score	2
Claim impact	SUPPORTIVE BUT RETROSPECTIVE
Caveat / interpretation	Cross-conditional evidence is exactly what the language phase is meant to organize.
Source	Open source

LIT-014 — Environment (P04)

Simulation / sample	TNG50/100/300
Scope	Satellite population
Published relation	Satellite SHMR scatter exceeds central scatter and the shift is produced mostly by outside-in tidal stripping; aperture choice changes the measured departure.
Numeric anchor	Scatter larger by up to 0.8 dex
Target record	Dynamical mass profile / SHMR
Conditioning variables	Aperture; stripping depth; environment
External status	Published TNG result
UM mapping	Nested measurement Boundaries expose different retained layers of the same object; outer information is preferentially removed.
Generic baseline / missing structure	A fixed-aperture scalar catalogue obscures the mechanism.
Compatibility	Strong
Distinctive?	YES
Score	2
Claim impact	SUPPORTIVE BUT RETROSPECTIVE
Caveat / interpretation	Very strong scale-indexed Boundary example.
Source	Open source

LIT-015 — Environment (P05)

Simulation / sample	IllustrisTNG
Scope	z≤0.5; M*≥10^10.5 Msun
Published relation	Massive centrals and satellites are quenched at similar high rates largely regardless of environment, consistent with internal AGN-driven quenching.
Numeric anchor	~70–90% quenched
Target record	Quenched state
Conditioning variables	Stellar mass; AGN channel; environment
External status	Published TNG result
UM mapping	The active causal channel changes with scale and mass: the same output class can be reached through an internal rather than environmental path.
Generic baseline / missing structure	Environment-only language fails for massive systems.
Compatibility	Strong
Distinctive?	YES
Score	2
Claim impact	SUPPORTIVE BUT RETROSPECTIVE
Caveat / interpretation	Prevents UM from reducing every result to external Boundary action.
Source	Open source

LIT-016 — Environment (P05)

Simulation / sample	IllustrisTNG
Scope	M*≲10^10 Msun in groups/clusters
Published relation	Low-mass centrals are rarely quenched in isolation but become commonly quenched after entering massive hosts.
Numeric anchor	Quenched fraction rises to ~80%
Target record	Quenched state
Conditioning variables	Host mass; membership; stellar mass
External status	Published TNG result
UM mapping	Environmental Boundary conditions dominate retention for low-mass systems, while the mass-conditioned channel changes at higher mass.
Generic baseline / missing structure	Mass-only and environment-only descriptions each fail without the interaction.

Compatibility	Strong
Distinctive?	YES
Score	2
Claim impact	SUPPORTIVE BUT RETROSPECTIVE
Caveat / interpretation	Strong conditional language test.
Source	Open source

LIT-017 — Environment (P05)

Simulation / sample	IllustrisTNG
Scope	Low-mass satellites in massive hosts
Published relation	Satellites accreted more than roughly 4–6 Gyr ago are almost entirely passive.
Numeric anchor	Environmental-quenching timescale ≤4–6 Gyr
Target record	Quenched state
Conditioning variables	Time since infall; host mass
External status	Published TNG result
UM mapping	Processing age on retained history is essential; current environment alone does not encode elapsed exposure.
Generic baseline / missing structure	Present-day density or membership lacks the causal clock.
Compatibility	Strong
Distinctive?	YES
Score	2
Claim impact	SUPPORTIVE BUT RETROSPECTIVE
Caveat / interpretation	Direct support for keeping processing age separate.
Source	Open source

LIT-018 — Environment (P05)

Simulation / sample	IllustrisTNG
Scope	z=0 group/cluster satellites
Published relation	A substantial fraction of quenched satellites were already quenched before entering their current host.

Numeric anchor	~30% pre-quenched; many 4–10 Gyr ago
Target record	Quenching history
Conditioning variables	Prior host; preprocessing; formation path
External status	Published TNG result
UM mapping	The current Boundary does not contain the complete processing history; nested past environments remain causally relevant.
Generic baseline / missing structure	Current-host-only classification cannot reconstruct the record.
Compatibility	Strong
Distinctive?	YES
Score	2
Claim impact	SUPPORTIVE BUT RETROSPECTIVE
Caveat / interpretation	Strong retained-history and nested-boundary result.
Source	Open source

LIT-019 — Structure (P07)

Simulation / sample	TNG100/300
Scope	Massive groups/clusters at $z \leq 1$
Published relation	In the richest clusters, roughly half the stellar mass is in satellites and half in the central plus diffuse intracluster component; the ICL share depends on the central definition.
Numeric anchor	ICL ~20–40% of total stellar mass
Target record	Stellar mass partition
Conditioning variables	Object definition; aperture; host mass
External status	Published TNG result
UM mapping	Whole/part/remainder accounting changes with the declared identity Boundary; diffuse material is not absent, only assigned to a different class.
Generic baseline / missing structure	One fixed central-galaxy definition cannot preserve the full partition.
Compatibility	Strong
Distinctive?	YES
Score	2
Claim impact	SUPPORTIVE BUT RETROSPECTIVE

Caveat / interpretation	A clear nested-object and classification-boundary example.
Source	Open source

LIT-020 — Structure (P07)

Simulation / sample	TNG100/300
Scope	z=0 groups/clusters
Published relation	Central stellar mass within 30 kpc is tightly related to host halo mass.
Numeric anchor	Mstar(<30 kpc) ∝ M500c^0.49; scatter ~0.12 dex
Target record	Stellar mass
Conditioning variables	Halo mass; 30 kpc aperture
External status	Published TNG result
UM mapping	A stable ensemble relation exists at a declared aperture.
Generic baseline / missing structure	Halo-mass-only regression already captures the stated relation.
Compatibility	Partial
Distinctive?	NO
Score	1
Claim impact	COMPATIBLE / NON-DISTINCTIVE
Caveat / interpretation	Compatible but not distinctive.
Source	Open source

LIT-021 — Structure (P07)

Simulation / sample	TNG100/300
Scope	z=0 groups/clusters
Published relation	Massive-galaxy half-mass radius is tightly related to halo mass.
Numeric anchor	r_half ∝ M500c^0.53; scatter ~0.16 dex
Target record	Stellar size
Conditioning variables	Halo mass; size definition
External status	Published TNG result

UM mapping	A stable scale-declared structural field emerges from many local records.
Generic baseline / missing structure	Halo-mass scaling is already a generic description.
Compatibility	Partial
Distinctive?	NO
Score	1
Claim impact	COMPATIBLE / NON-DISTINCTIVE
Caveat / interpretation	Compatible but non-distinctive.
Source	Open source

LIT-022 — Clustering (P08)

Simulation / sample	TNG100/300
Scope	Nonlinear scales
Published relation	Baryonic physics modifies the total matter field and suppresses the power spectrum over a finite scale range.
Numeric anchor	Up to ~20% damping for $k \lesssim 10$ h/Mpc
Target record	Matter power spectrum
Conditioning variables	Scale; baryonic component
External status	Published TNG result
UM mapping	Collective fields are scale-dependent ensemble summaries; changing retained components changes the coarse field without adding a new substance.
Generic baseline / missing structure	Standard baryonic-correction language already explains the effect.
Compatibility	Partial
Distinctive?	NO
Score	1
Claim impact	COMPATIBLE / NON-DISTINCTIVE
Caveat / interpretation	Strong conceptual fit, weak distinctiveness.
Source	Open source

LIT-023 — Clustering (P08)

Simulation / sample	TNG100/300
Scope	High redshift to z=0
Published relation	The nonlinear stellar-mass two-point correlation function is close to a power law and approximately invariant over a wide temporal range.
Numeric anchor	Near power-law; approximate time invariance
Target record	Two-point correlation
Conditioning variables	Scale; tracer; redshift
External status	Published TNG result
UM mapping	Stable ensemble statistics support a familiar collective descriptor across epochs.
Generic baseline / missing structure	Ordinary correlation-function language already captures it.
Compatibility	Partial
Distinctive?	NO
Score	1
Claim impact	COMPATIBLE / NON-DISTINCTIVE
Caveat / interpretation	Compatibility evidence only.
Source	Open source

LIT-024 — Clustering (P08)

Simulation / sample	TNG100/300
Scope	z=0 to z>1; BAO scales
Published relation	Clustering slope and tracer bias vary with epoch and scale, including small correctable acoustic-peak shifts.
Numeric anchor	gamma ~1.8 at z=0 to ~1.6 at z~1; BAO shift ~5%
Target record	Correlation slope / tracer bias
Conditioning variables	Epoch; tracer channel; scale
External status	Published TNG result
UM mapping	A record must declare scale, channel, and processing epoch; one global clustering label is insufficient.
Generic baseline / missing structure	A single clustering scalar or timeless bias parameter misses the conditional structure.

Compatibility	Strong
Distinctive?	YES
Score	2
Claim impact	SUPPORTIVE BUT RETROSPECTIVE
Caveat / interpretation	Distinctive as a language requirement, not as a new physical prediction.
Source	Open source

LIT-025 — Clustering (P08)

Simulation / sample	TNG100/300
Scope	z≈0.1; red galaxies 10^9–10^10 Msun/h^2
Published relation	TNG shows a mild excess in red-galaxy clustering relative to SDSS in a stated mass interval.
Numeric anchor	Mild published excess
Target record	Galaxy clustering
Conditioning variables	Colour; stellar mass; redshift
External status	TNG-vs-observation tension
UM mapping	Retained as an external simulation tension rather than counted as language success.
Generic baseline / missing structure	No interpretive framework removes the quantitative mismatch.
Compatibility	Neutral
Distinctive?	NO
Score	0
Claim impact	TNG TENSION — NOT UM FAILURE
Caveat / interpretation	Keeps the comparison ledger open-world and non-confirmatory.
Source	Open source

LIT-026 — Compute / Access (P01)

Simulation / sample	TNG public release
Scope	All flagship runs
Published relation	The complete release is enormous, but the project provides halo/subhalo catalogues, an API, exploratory tools, and hosted Jupyter analysis to avoid full-volume downloads.

Numeric anchor	~1.1 PB; 100 redshifts; API/JupyterLab
Target record	Data-access pathway
Conditioning variables	Compute; storage; requested analysis scale
External status	Access constraint / method
UM mapping	The full prospective catalogue test remains valid but is deferred; published summaries are an interim evidence layer, not a replacement.
Generic baseline / missing structure	Not a scientific baseline comparison.
Compatibility	Neutral
Distinctive?	NO
Score	0
Claim impact	NEUTRAL
Caveat / interpretation	Justifies the staged test ladder under the user's compute constraint.
Source	Open source

9.11 Literature scorecard

Measure	Result
Primary papers	9
Evidence statements	26
Strong retrospective mappings	14
Partial/compatible mappings	8
Neutral statements	4
TNG-versus-observation tensions	3
UM tensions or contradictions	0
Final verdict	Language survives; superiority untested.

PART X — SPARC REAL-GALAXY PILOT

10. Why this pilot was run

The SPARC pilot answered a practical question: can Universopedia/UM be tested on genuine object-level data without petabyte storage or institutional hardware? The answer was yes. It also exposed the difference between testing a representation and testing a new physical mechanism.

10.1 Frozen execution

Protocol element	Executed choice
Data	12 real SPARC galaxies; 270 resolved radial measurements.
Evaluation	Leave one entire galaxy out, train on the other 11, repeat for all 12.
Baryonic components	Gas, disk, and bulge velocity contributions from published mass-model files.
Compiler shares	a_L from gas/update contribution; a_M = 1 − a_L from stellar retention contribution.
UM coordinates	$W_L = a_L^2$; $W_B = 2a_La_M$; $W_M = a_M^2$.
Comparators	Baryons only, fixed RAR, generic linear inputs, generic quadratic inputs.

10.2 Results

Model	Galaxy-balanced RMSE dex	Point RMSE dex	Galaxy-balanced V RMSE km/s	Galaxy-balanced V MAPE
RAR-fixed	0.2404	0.2582	17.15	32.10%
Raw-1	0.3154	0.3001	27.34	33.02%
UM-Q	0.3273	0.3133	30.60	33.77%
Raw-2	0.3273	0.3133	30.60	33.77%
Baryons-fixed	0.5275	0.5559	43.33	41.89%

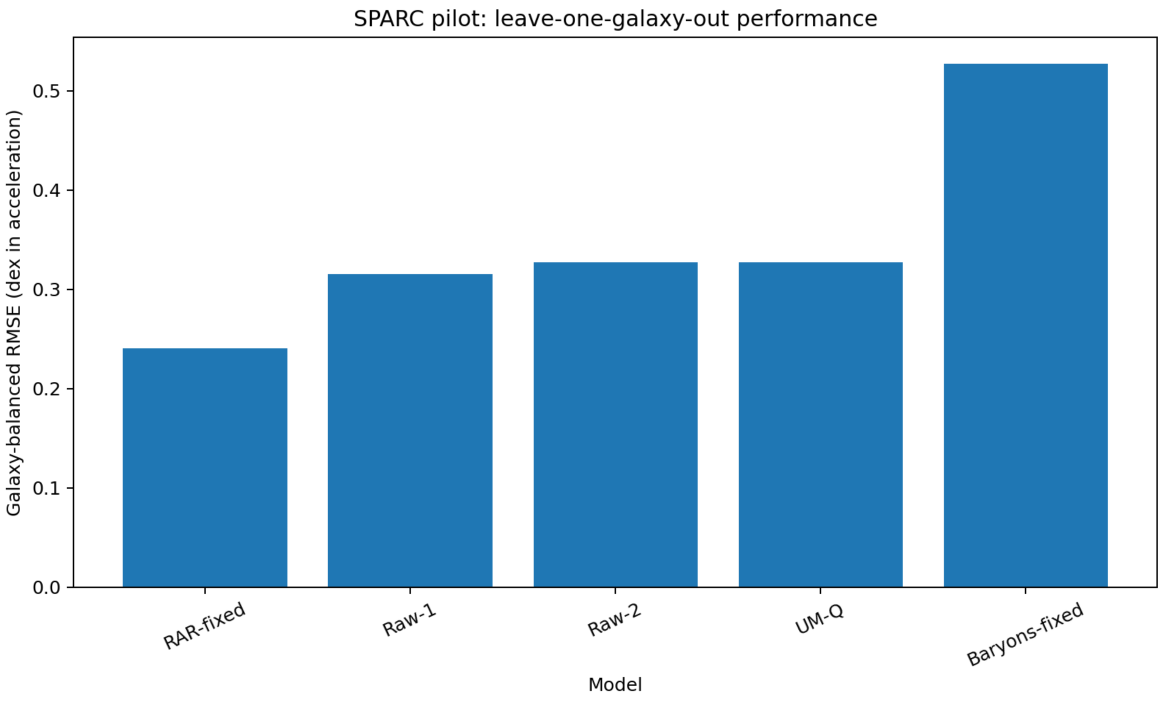

Figure 1. Leave-one-galaxy-out model comparison. Lower acceleration RMSE is better.

10.3 Algebraic equivalence result

$$W\_B = 2a\_L - 2a\_L^2$$
$$W\_M = 1 - 2a\_L + a\_L^2$$
$$\text{span}\{1,W\_B,W\_M\} = \text{span}\{1,a\_L,a\_L^2\}$$

UM-Q and the generic Raw-2 quadratic baseline produced the same predictions to numerical precision. This is a valid null result: the chosen channel coordinates are a coordinate transformation of the same quadratic information, not an extra predictive degree of freedom.

SPARC VERDICT

The compiler executed successfully on real records, improved over baryons alone, did not beat the established radial-acceleration relation, and supplied no distinctive evidence for a new galactic mechanism.

This pilot did not test the conditional silver-ratio galactic branch because no UM-derived radial operator or circular-velocity observable had been supplied. It tested the interpretive compiler, not a new force law.

PART XI — GOVERNANCE AND FUTURE USE

11. Conservative language growth

Universopedia is open-world: unfamiliar records may require new classes. Growth is conservative only when the new class is source-traceable, operationally defined, reproducible across appropriate records, compatible with existing validated classes, and explicit about its own validity Boundary and uncertainty.

15. Register the candidate record without forcing a familiar label.
16. Show that the structure is preserved in the data and not generated by preprocessing.
17. Define the new class operationally and identify distinguishing records.
18. Compare against existing classes and generic baselines.
19. Test transfer across instruments, resolutions, systems, or epochs when applicable.
20. Version the decoder and preserve old classifications for audit.
21. Promote the class only after the evidence grade is stated and the source ledger is complete.

11.1 Standard empirical-test template

Field	Requirement
Protocol ID	Unique identifier and version.
Scientific layer	Core observational compiler, ensemble interpretation, or optional dynamics.
Mechanism	Physical process claimed, or "downstream interpretation only."
Adapter inputs	Exact fields available before target inspection.
Withheld targets	Records protected from the adapter.
Data identity	Source, version, checksum, sample definition, units.
Adapter identity	Source code, checksum, parameter freeze.
Partition	Leakage-resistant train/validation/test rule.
Baselines	Null, information-matched, generic physical, and incremental-value models.
Transfer	Resolution, volume, instrument, system, or epoch transfer without retuning.
Metrics	Loss, uncertainty, calibration, effect size, complexity penalty.
Null tests	Shuffle, order, rescaling, leakage, and implementation controls.
Failure condition	What outcome falsifies the mechanism or defeats distinctiveness.
Publication rule	Null and baseline-equivalent results remain results.

11.2 Distinctiveness rule

$$
\begin{gathered}
\text{coverage} \neq \text{transfer} \\
\text{transfer} \neq \text{withheld prediction} \\
\text{withheld prediction} \neq \text{baseline advantage} \\
\text{baseline advantage} \neq \text{validated new dynamics}
\end{gathered}
$$

Universopedia may be valuable as an explanatory and organizational language even when it does not outperform a specialist fit. A claim of empirical advantage requires a frozen compiler that transfers, predicts protected records, and exceeds fair generic baselines. A claim of new physics additionally requires a physical mechanism, observable, calibration, and likelihood pipeline.

11.3 What the project has established

- A complete formal distinction between records, established inference, UM reconstruction, Transmission limits, and Familiarity gaps.
- A scale- and channel-dependent account of effective objects and actual Boundaries.
- A four-clock age language that prevents propagation delay, formation, material history, and processing history from being collapsed.
- An ensemble rule explaining when local records support field language.
- A complete periodic language family and representative cross-scale atlas.
- A source-traceable observational mathematics catalogue and complexity-profile system.
- A cosmology ledger that preserves coverage, open tensions, and lack of independent UM prediction honestly.
- A prospective TNG protocol with leakage controls and a completed retrospective literature phase.
- A real SPARC execution showing that ordinary hardware can run object-level tests and that a baseline-equivalent result is publishable evidence about the compiler.

11.4 What remains open

- Recovery or formal refreezing of the exact TNG adapter and authenticated catalogue execution.
- A physical derivation of any UM-specific observable before claiming a force, velocity law, detector shift, or coupling.
- Cross-sector evidence that the same physical anchors govern multiple domains without sector-specific retuning.
- A distinctive transfer or withheld-prediction advantage beyond generic coarse-graining.
- A solved optional dynamics branch that survives existing laboratory, Solar-System, astrophysical, and cosmological constraints.

FINAL UNIVERSOPEDIA STATUS
FORMAL OBSERVATIONAL LANGUAGE CLOSED / FULL CORE LEDGER CATEGORIZED / COSMOLOGY COVERAGE COMPLETE / TNG PROSPECTIVE TEST DEFERRED / TNG LITERATURE COMPATIBILITY COMPLETE / SPARC REAL-DATA NULL RESULT RETAINED / PHYSICAL DISTINCTIVENESS OPEN.

APPENDIX A — SOURCE INVENTORY

Core, cosmology, and TNG sources are separated from local UM artifacts. External sources support the established layer; local artifacts govern definitions, interpretations, protocols, and project status.

A.1 Core source ledger

ID	Title	Organization/authors	Type	Scope	Accessed	Local note
SRC-001	Periodic Table of the Elements	NIST	Official evaluated reference	Atomic properties and current table	2026-07-14	
SRC-002	Periodic Table of Elements	IUPAC	Official reference	Names and standard atomic-weight framework	2026-07-14	
SRC-003	Basics of Space Flight: Solar System	NASA/JPL	Official educational reference	Representative planetary ratios and channels	2026-07-14	
SRC-004	Solar System Exploration	NASA Science	Official reference	Solar-system object context	2026-07-14	
SRC-005	Stars	NASA Science	Official synthesis	Stellar lifecycle classes	2026-07-14	
SRC-006	Sun Facts	NASA Science	Official reference	Solar age, class and scale	2026-07-14	
SRC-007	Galaxies	NASA Science	Official synthesis	Galaxy classes and Milky Way context	2026-07-14	
SRC-008	Messier 31	NASA Hubble	Official reference	Andromeda distance and observational context	2026-07-14	
SRC-009	Messier 87	NASA Hubble	Official reference	M87 distance, stars, clusters and jet	2026-07-14	
SRC-010	First M87 EHT Results VI	EHT Collaboration	Primary paper	M87* ring scale and mass inference	2026-07-14	arXiv
SRC-011	First Sagittarius A* EHT Results I	EHT Collaboration	Primary paper	Sgr A* ring and horizon-scale interpretation	2026-07-14	arXiv
SRC-012	First Sagittarius A* EHT Results IV	EHT Collaboration	Primary paper	Variability, morphology and calibrated mass	2026-07-14	arXiv
SRC-013	Cygnus X-1 contains a 21-solar-mass black hole	Miller-Jones et al.	Primary paper	Distance and black-hole mass	2026-07-14	arXiv
SRC-014	GW150914 properties	LIGO/Virgo Collaboration	Primary paper	Binary and remnant inference	2026-07-14	arXiv

ID	Title	Organization/authors	Type	Scope	Accessed	Local note
SRC-015	Fundamental Physical Constants	NIST CODATA	Official evaluated reference	Constants used in observable equations	2026-07-14	
SRC-016	How We Find and Characterize Exoplanets	NASA Science	Official synthesis	Transit and characterization channels	2026-07-14	
SRC-017	NICER PSR J0740+6620 result	NASA HEASARC	Official mission summary	Neutron-star radius and mass inference	2026-07-14	
SRC-018	Small Magellanic Cloud	NASA Science	Official reference	Dwarf-galaxy observational context	2026-07-14	
SRC-019	Hubble Views Starburst Galaxy M82	NASA Science	Official reference	Starburst-galaxy channels	2026-07-14	
UM-029	Nature-Dependent Boundary Hierarchy	Unified Mechanics	Local interpretive artifact	Environment-gated candidate branch	2026-07-14	outputs/UM_029_Nature_Dependent_Boundary_Hierarchy.md
UM-030	Actual Boundary and Jupiter Example	Unified Mechanics	Local interpretive artifact	Scale-indexed causal cut and four clocks	2026-07-14	outputs/UM_030_Actual_Boundary_Causal_Scale_and_Jupiter_Example.md
UM-031	Familiarity Kernel and Observational Language	Unified Mechanics	Local formal artifact	Transmission/Familiarity split	2026-07-14	outputs/UM_031_Familiarity_Kernel_and_Observational_Language.md
UM-032	Cross-Scale Nature Familiarity Compiler	Unified Mechanics	Local formal artifact	Effective objects, hierarchy and compiler schema	2026-07-14	outputs/UM_032_Cross_Scale_Nature_Familiarity_Compiler.md

A.2 Cosmology source ledger

ID	Title	Organization/authors	Type	Scope	Accessed	Used by
COSRC-001	Planck 2018 results VI: Cosmological parameters	Planck Collaboration	Primary collaboration paper	CMB parameters, H0, densities, ns, tau, Neff, neutrino mass	https://arxiv.org/abs/1807.06209	2026-07-14
COSRC-002	ACT DR6 Power Spectra, Likelihoods and LambdaCDM Parameters	ACT Collaboration	Primary collaboration paper	Independent CMB spectra and joint constraints	https://arxiv.org/abs/2503.14452	2026-07-14
COSRC-003	DESI DR2 Results II: BAO and Cosmological Constraints	DESI Collaboration	Primary collaboration paper	Galaxy/quasar BAO, neutrino mass, dark-energy comparison	https://arxiv.org/abs/2503.14738	2026-07-14
COSRC-004	DESI DR2 Results I: Lyman-alpha Forest BAO	DESI Collaboration	Primary collaboration paper	High-redshift intergalactic BAO	https://arxiv.org/abs/2503.14739	2026-07-14

ID	Title	Organization/authors	Type	Scope	Accessed	Used by
COSRC-005	DES five-year supernova cosmology	DES Collaboration	Primary collaboration paper	Type Ia supernova expansion history	https://arxiv.org/abs/2401.02929	2026-07-14
COSRC-006	Comprehensive local H0 measurement	SH0ES Team	Primary collaboration paper	Cepheid-SN Ia distance ladder	https://arxiv.org/abs/2112.04510	2026-07-14
COSRC-007	Dark and bright standard-siren H0 after O4a	Bom et al.	Primary analysis	Gravitational-wave distance scale	https://arxiv.org/abs/2404.16092	2026-07-14
COSRC-008	DES Year 3 galaxy clustering and weak lensing	DES Collaboration	Primary collaboration paper	Late-time structure and S8	https://arxiv.org/abs/2105.13549	2026-07-14
COSRC-009	DES Y3 cluster abundances, weak lensing and clustering	DES Collaboration	Primary collaboration paper	Cluster-calibrated late growth	https://arxiv.org/abs/2503.13632	2026-07-14
COSRC-010	eROSITA all-sky cluster cosmology	eROSITA Collaboration	Primary collaboration paper	X-ray cluster counts with lensing calibration	https://arxiv.org/abs/2402.08458	2026-07-14
COSRC-011	Planck 2018 results IX: Primordial non-Gaussianity	Planck Collaboration	Primary collaboration paper	CMB bispectrum constraints	https://arxiv.org/abs/1905.05697	2026-07-14
COSRC-012	BICEP/Keck XIII primordial gravitational waves	BICEP/Keck Collaboration	Primary collaboration paper	Primordial tensor upper limit	https://arxiv.org/abs/2110.00483	2026-07-14
COSRC-013	The 2024 BBN baryon abundance update	Schoneberg	Primary synthesis and calculation	Light-element baryon and Neff constraints	https://arxiv.org/abs/2401.15054	2026-07-14
COSRC-014	HERA Phase I reionization 21 cm upper limits	HERA Collaboration	Primary collaboration paper	Epoch-of-reionization 21 cm power	https://arxiv.org/abs/2108.02263	2026-07-14
COSRC-015	CMB spectral distortions revisited with FIRAS	Bianchini and Fabbian	Primary reanalysis	CMB mu-distortion upper limit	https://arxiv.org/abs/2206.02762	2026-07-14
COSRC-016	MICROSCOPE final equivalence-principle result	Touboul et al.	Primary mission paper	Composition-dependent acceleration null test	https://arxiv.org/abs/2209.15487	2026-07-14

ID	Title	Organization/authors	Type	Scope	Accessed	Used by
COSRC-017	Constraining symmetron dark energy using atom interferometry	Chiow and Yu	Primary analysis	Laboratory screening constraints	https://arxiv.org/abs/1911.00441	2026-07-14
COSRC-018	Working-universe simulation reinterpretation	Unified Mechanics	Local implementation record	CAMB/Planck downstream interpreter	outputs/UM_038_Working_Universe_Simulation_Reinterpretation.md	2026-07-14
COSRC-019	Record-to-field ensemble closure	Unified Mechanics	Local formal artifact	Single-record and ensemble distinction	outputs/UM_039_Record_to_Field_Ensemble_Closure.md	2026-07-14
COSRC-020	IllustrisTNG public data release	Nelson et al.	Primary collaboration paper	Hydrodynamic structure-catalogue context	https://arxiv.org/abs/1812.05609	2026-07-14
COSRC-021	IllustrisTNG Web API Documentation	IllustrisTNG Collaboration	Official data documentation	Authentication, queries, downloads and API access	https://www.tng-project.org/data/docs/api/	2026-07-14
COSRC-022	IllustrisTNG Data Specifications	IllustrisTNG Collaboration	Official data documentation	Group and subhalo field definitions, units and merger-tree fields	https://www.tng-project.org/data/docs/specifications/	2026-07-14

A.3 TNG literature sources

ID	Lead author	Year	Title	Domain	Simulation(s)	Identifier	Evidence IDs
P01	Nelson et al.	2019	The IllustrisTNG Simulations: Public Data Release	Data release / compute access	TNG50, TNG100, TNG300	arXiv:1812.05609	LIT-026
P02	Torrey et al.	2019	The evolution of the mass-metallicity relation in IllustrisTNG	Metallicity	Primarily TNG100	arXiv:1711.05261	LIT-001–LIT-004
P03	Diemer et al.	2019	Atomic and molecular gas in IllustrisTNG galaxies at low redshift	Gas	IllustrisTNG	arXiv:1902.10714	LIT-006–LIT-009
P04	Engler et al.	2021	The distinct stellar-to-halo mass relations of satellite and central galaxies	Environment / stripping	TNG50, TNG100, TNG300	doi:10.1093/mnras/staa3505	LIT-012–LIT-014
P05	Donnari et al.	2021	Quenched fractions in the IllustrisTNG simulations	Environment / quenching	IllustrisTNG	arXiv:2008.00005	LIT-015–LIT-018

ID	Lead author	Year	Title	Domain	Simulation(s)	Identifier	Evidence IDs
P06	Gupta et al.	2018	Chemical pre-processing of cluster galaxies over the past 10 billion years	Metallicity / environment	IllustrisTNG	arXiv:1801.03500	LIT-005
P07	Pillepich et al.	2018	First results: stellar mass content of groups and clusters	Structure	TNG100, TNG300	arXiv:1707.03406	LIT-019–LIT-021
P08	Springel et al.	2018	First results: matter and galaxy clustering	Clustering / collective fields	TNG100, TNG300	doi:10.1093/mnras/stx3304	LIT-022–LIT-025
P09	Engler et al.	2023	Satellites of Milky Way- and M31-like galaxies with TNG50	Gas / environment / age	TNG50	doi:10.1093/mnras/stad1357	LIT-010–LIT-011

APPENDIX B — GOVERNING ARTIFACT MAP

Artifact	Role
UM_036_Final_Consolidated_Edition_2026.pdf	Governing structural, observational, ensemble, and optional-physics status.
Unified_Mechanics_Main_Monograph_Addendum_2026.pdf	Insertion-ready monograph architecture and replacement language.
Unified_Mechanics_Compact_Monograph_2026.pdf	Compact reader-facing summary and claim grades.
Universopedia_Observational_Language_Ledger.xlsx	Core class dictionary, periodic table, cross-scale atlas, observable equations, profiles, sources, validation.
Universopedia_Cosmology_Comparison_2026.xlsx	29-probe cosmology ledger, tensions, comparison rubric, source and validation sheets.
Universopedia_TNG_Core_Test_Protocol.xlsx	Frozen prospective protocol, field map, run ledger, results grid, validation.
Universopedia_TNG_Literature_Comparison.xlsx	Prospective protocol plus 26-statement retrospective TNG comparison and scorecard.
UM_TNG_Core_Test_Starter_Package.zip	Harness, adapter contract, configuration, README, and self-test materials.
UM_SPARC_Pilot_Package.zip	Real-data pilot workbook, report, code, raw data, predictions, metrics, and figures.
UM_Mechanism_Derivation_Audit_2026-07-14.md	Audit of genuine derivation, conditional physical mechanism, reinterpretation, and remaining anchors.
Unified_Mechanics_Claude_Master_Document_Handoff_2026-07-14.docx	Editorial authority for the final Claude-assembled master document.
Universopedia_Formal_Complete_Edition_2026.docx	Standalone complete formal Universopedia standard and categorized ledger.

APPENDIX C — CLAUDE MASTER-DOCUMENT INTEGRATION INSTRUCTION

INSTRUCTION TO CLAUDE

Treat this Universopedia document as the governing formal statement of the observational-language layer. Do not compress it into a few paragraphs or merge Record, Established Inference, UM Reconstruction, Transmission, and Familiarity. Preserve the mechanism-before-measurement rule near the beginning of the master work and apply it to every empirical claim.

22. Introduce Universopedia after the structural core and before any claim of physical prediction.

23. State the full typed chain from physical mechanism to record and inference.
24. Retain the complete class dictionary, kernels, Boundary definition, four clocks, ensemble-field criterion, complexity profile, and validation gates.
25. Present the periodic table as the first complete physical language family and preserve its guardrails.
26. Use the planet, star, galaxy, and black-hole entries as worked examples of channel-limited description, not as decorative summaries.
27. Preserve all 22 observable equations with their direct inputs, inferred outputs, validity Boundaries, and grades.
28. Describe the cosmology comparison as a coverage audit with no independent UM prediction or withheld win.
29. Describe the TNG literature phase as retrospective compatibility only and the catalogue test as deferred pending exact adapter identity and data execution.
30. Describe the SPARC pilot as a successful real-data execution and a genuine null/baseline-equivalence result, not as a failure of the entire framework or a test of a nonexistent UM rotation law.
31. Never write that Universopedia measures the whole physical world. It formalizes how particular mechanisms and channels produce particular records from which bounded descriptions become possible.

CLOSING STATEMENT

Universopedia does not claim that the universe is a book already written in human categories. It is a disciplined method for showing how the world leaves partial records, how established models reconstruct stable quantities from those records, how language classes become familiar across scales, where physical information was never transmitted, and where a proposed mechanism must still risk failure. Its strength is not omniscience. Its strength is that it refuses to confuse a record with a world, a description with a mechanism, or a mechanism with a measurement before the complete chain exists.